Camassa–Holm peakons under momentum transport noise. Part II: Multipeakon stability and continuous dependence

Lei Zhang[1*] and Xiuting Li[2]

[1]School of Mathematics and Statistics, Hubei Key Laboratory of Engineering Modeling and Scientific Computing, Huazhong University of Science and Technology, Luoyu Road 1037, Wuhan, 430074, China.
[2]College of Informatics, Huazhong Agricultural University, No. 1 Shizishan Street, Hongshan District, Wuhan, Hubei, 430070, China.

*Corresponding author(s). E-mail(s): lei_zhang@hust.edu.cn;
Contributing authors: xtingli@mail.hzau.edu.cn;

Abstract

Part I [22] (L. Zhang and X. Li, *Camassa–Holm peakons under momentum transport noise. Part I: Global well-posedness and probabilistic stability*, companion manuscript, 2026) established global well-posedness for nonnegative finite momentum measures and probabilistic stability of the positive single-peakon family under Stratonovich momentum transport noise. Here we prove probabilistic stability of the positive multipeakon set and $\boldsymbol{H^1}$ continuous dependence at a specified multipeakon driven by the same Brownian motions. Spectral tails of the Green operator yield positive atomic approximations with a sharp exponent in the $\boldsymbol{H^1}$ approximation inequality. Relative exterior-power estimates yield exit probabilities and lower bounds for mean exit times without a uniform bound on the perturbations' total momentum. Comparing localized momentum components shows that initial $\boldsymbol{H^1}$ convergence to a fixed multipeakon implies convergence in every finite $\boldsymbol{L^p}$ norm of the supremum in time of the $\boldsymbol{H^1}$ error on each finite interval, also without such a bound. The resulting modulus is explicit for spatially varying noise, with a sharp one-half exponent for spatially constant noise. Local estimates also recover the weights and positions of nearby finite multipeakon profiles.

Keywords: stochastic Camassa–Holm equation, multipeakon, nonnegative momentum measure, spectral estimates, continuous dependence, exit time

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1 Introduction

1.1 Formulation of the problem

The Camassa–Holm (CH) equation was derived as a model for shallow water waves and admits peaked solitary waves [6, 7]. Its dispersionless form on the real line is

$$m_t + um_x + 2u_x m = 0, \qquad m = u - u_{xx}, \qquad u = G * m, \quad G(x) = \tfrac{1}{2}e^{-|x|}.$$

Nonnegative finite momentum measures provide a natural class of global weak solutions [10]. This class includes positive multipeakons,

$$m_t = \sum_{i=1}^{n} p_i(t)\delta_{q_i(t)}, \qquad u_t = \sum_{i=1}^{n} p_i(t)G(\cdot - q_i(t)), \qquad p_i(t) > 0,$$

as well as velocities with diffuse momentum. Here p_i is a momentum weight, so an isolated profile p_iG has height $p_i/2$. The corners at the crests correspond to atoms of the momentum.

The stability theory of these waves is closely related to the conservation and spectral properties of deterministic CH. Conserved quantities control perturbations of single peakons and sufficiently separated positive peakon trains [11, 16], while the spectral formulation relates multipeakon dynamics to strings and Stieltjes moment problems [2–4]. These connections lead us to study how the spectral structure can measure and propagate proximity to positive multipeakons under stochastic momentum transport.

The stochastic model considered here arises from Holm's variational formulation [12, 18, 19]. For the CH Lagrangian,

$$\ell(u) = \frac{1}{2}\int_{\mathbb{R}} (u^2 + u_x^2)\,\mathrm{d}x, \qquad m = \frac{\delta \ell}{\delta u} = u - u_{xx},$$

the Euler–Poincaré equation transports the momentum as a one-form density. Holm introduces Stratonovich noise through the transport differential

$$\mathrm{d}\chi_t = u_t\,\mathrm{d}t + \varepsilon \sum_{k=1}^{N} \xi_k \circ \mathrm{d}W_t^k.$$

For smooth velocities, the one-dimensional Lie derivative of the momentum density is

$$\mathcal{L}_v(m\ \mathrm{d}x \otimes \mathrm{d}x) = (vm_x + 2v_x m)\ \mathrm{d}x \otimes \mathrm{d}x.$$

Consequently, the stochastic transport law $\mathrm{d}(m\ \mathrm{d}x \otimes \mathrm{d}x) + \mathcal{L}_{\mathrm{d}\chi_t}(m\ \mathrm{d}x \otimes \mathrm{d}x) = 0$ gives the equation below, with the noise sign included in the prescribed fields:

$$\mathrm{d}m + (um_x + 2u_x m)\,\mathrm{d}t + \varepsilon \sum_{k=1}^{N} (\xi_k m_x + 2\xi_k' m) \circ \mathrm{d}W^k = 0, \qquad m = u - u_{xx}. \quad (1.1)$$

Thus the factor 2 in the stochastic term follows from the same momentum geometry as the deterministic drift. Throughout, $W^1, \dots, W^N$ are independent Brownian motions, $\xi_k \in C_b^3(\mathbb{R})$ are deterministic, and $0 \le \varepsilon \le \varepsilon_*$. For each fixed ε, replacing ξ_k by $\widetilde{\xi}_k = \varepsilon \xi_k$ incorporates the amplitude into the fields, as in Holm's notation. Keeping ε explicit records the dependence of the estimates on noise strength and includes deterministic CH at $\varepsilon = 0$. Definition 2.1 extends this formulation to deterministic nonnegative finite initial momentum measures.

Positive atomic initial data evolve within the multipeakon ansatz under (1.1) [12, 19]. To study perturbations in the full momentum class, we use the global pathwise solution theory established in Part I [22], where the stability of the single-peakon family is also proved. The present paper develops the stability theory for positive multipeakons, allowing general nonnegative finite momentum perturbations whose velocity error is small in H^1. This permits diffuse components with arbitrarily large total momentum, with the initial energy error serving as the basis of the comparison.

We consider both proximity to the multipeakon family and dependence on a specified multipeakon trajectory. Writing $\mathcal{P}_n$ for the profiles with at most n positive peaks and v_t for a fixed multipeakon driven by the same Brownian motions as u_t, the corresponding errors on $[0, T]$ are

$$\sup_{t \le T} \operatorname{dist}_{H^1}(u_t, \mathcal{P}_n), \qquad \sup_{t \le T} \|u_t - v_t\|_{H^1}.$$

For the family comparison, we construct positive atomic approximations from a spectral rank defect, with a sharp approximation exponent. Relative estimates for exterior powers of the Green operator then yield exit-probability bounds from H^1 neighborhoods of $\mathcal{P}_n$ and lower bounds for mean exit times, without peak separation or a lower weight bound.

For the trajectory comparison, localized momentum estimates and control of the remaining energy give continuous dependence at every fixed positive multipeakon initial profile in $L^p(\Omega; C([0, T]; H^1))$ for every finite $p \ge 1$ and $T > 0$. Neither comparison requires a uniform bound on the perturbation's total momentum. We obtain an explicit modulus for spatially varying noise and, for spatially constant noise or $\varepsilon = 0$, a one-half exponent for sufficiently small initial errors; this exponent is sharp over the multipeakon class. For reference profiles with positive separation and a positive lower weight bound, local estimates further recover the weights and positions of nearby profiles with at most the same number of positive peaks. These results describe the stability of the multipeakon family and the continuity of its stochastic dynamics, with precise statements in Section 1.3.

1.2 Related works

Constantin and Molinet [10] developed the deterministic solution theory with nonnegative momentum. Constantin and Strauss [11] proved single-peakon orbital stability. El Dika and Molinet [16, Theorem 1.1 and Corollary 1.1] proved stability of ordered, sufficiently separated peakon trains and deduced stability of the multipeakon set near any prescribed positive multipeakon initial datum. The latter conclusion holds for all

times and arbitrary strictly ordered positions, for nonnegative perturbation momentum with uniformly bounded total mass. Our stochastic set estimates hold on finite intervals without this uniform momentum bound.

Continuous dependence for conservative CH has been studied through transportation and Lipschitz metrics [5, 8, 17]. These distances concern the conservative state, which can include a concentrated energy measure. Our comparison at a specified positive multipeakon uses the ordinary H^1 velocity error. The spectral structure of multipeakons is connected to Stieltjes moment problems, strings and Jacobi matrices [2–4]; extensions to measure coefficients and conservative states were developed in [13–15]. This spectral theory provides the setting for the positive approximation result below.

For stochastic CH, the variational singular-solution framework and the evolution of peakons were developed in [12, 19]. Albeverio et al. [1] proved local strong well-posedness for momentum in $H^{2,q}(\mathbb{R})$, $1 < q < \infty$, with one Brownian driver and a first-order convection operator that includes full momentum transport. Their Doss–Sussman transformation and Kato argument assume C_b^4 regularity of the transport field. Dependence on initial data in lower Sobolev norms, up to a stopping time controlling higher norms, was studied for Itô multiplicative noise in [20]. Chen et al. [9, Theorem 1.1] obtained modulation and exit estimates starting from an exact smooth solitary wave, under Marcus jump velocity transport noise for a linearly dispersive CH model. The present results concern arbitrary nonnegative measure perturbations of singular multipeakons under (1.1).

Part I [22] supplies the global pathwise solution theory for nonnegative finite momentum measures, including atomic data and data without spatial moments. It also proves single-peakon family stability through the quadratic and cubic integrals. Using that solution theory, we study stability of the multipeakon set and continuous dependence at a specified multipeakon. The main issue is to control these two errors using the initial velocity error when the perturbation's total momentum has no uniform bound. For the set distance, exterior powers replace the quadratic–cubic defect of Part I; for a specified trajectory, localized momentum and remainder energy estimates provide a separate comparison argument.

1.3 Main results

Let $\mathcal{M}_+(\mathbb{R})$ denote the space of nonnegative finite Radon measures on $\mathbb{R}$. We use the energy norm $\|f\|_{H^1}^2 = \int_{\mathbb{R}} (f^2 + f_x^2)\,\mathrm{d}x$. For $m \in \mathcal{M}_+(\mathbb{R})$, put

$$u_m = G * m, \qquad M(m) = m(\mathbb{R}), \qquad E(m) = \|u_m\|_{H^1}^2.$$

The total momentum is $M(m)$; no spatial moment condition is imposed. In the stochastic statements, m_t is the global solution with deterministic initial momentum $m_0 \in \mathcal{M}_+(\mathbb{R})$, on the prescribed Brownian probability space, and $u_t = G * m_t$. The weak formulation and the solution theorem from Part I are stated in Definition 2.1 and Theorem 2.2. All comparisons of specified solutions use the same Brownian motions.

For an integer $n \geq 1$, define

$$\mathcal{P}_n = \left\{ \sum_{a=1}^{s} p_a G(\cdot - q_a) : 1 \leq s \leq n,\ p_a > 0,\ q_a \in \mathbb{R} \right\},$$
$$d_n(m) = \operatorname{dist}_{H^1}(u_m, \mathcal{P}_n).$$

For $n = 1$, $\mathcal{P}_1 = \mathcal{P}$ in the notation of Part I [22]. Allowing the zero profile does not change this distance, and $d_n(0) = 0$. The index n counts peaks, independently of the number N of Brownian motions. Neither total momentum nor peak positions are prescribed in this comparison.

The positive trace-class operator

$$(T_m f)(x) = \int_{\mathbb{R}} e^{-|x-y|/2} f(y) m(\,\mathrm{d}y) \quad \text{on } L^2(m)$$

has eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$, counted with multiplicity and completed with zeros. For $r \geq 1$, put

$$F_r(m) = \sum_{i_1 < \cdots < i_r} \lambda_{i_1}^2 \cdots \lambda_{i_r}^2.$$

Thus F_r is the squared Hilbert–Schmidt norm of the rth exterior power of T_m. Section 3 justifies these properties for arbitrary $m \in \mathcal{M}_+(\mathbb{R})$. We use the convention $E^0 = 1$, including at $E = 0$.

Theorem 1.1 (Approximation by positive multipeakons) *For every finite nonnegative measure m and integer $n \geq 1$,*

$$d_n(m)^{4n} \leq c_n E(m)^{n-1} F_{n+1}(m), \qquad c_n = 8^{n-1}\big((n+1)!\big)^2. \tag{1.2}$$

In particular,

$$d_2(m)^8 \leq 288\, E(m) F_3(m).$$

No total-momentum bound, support bound, amplitude lower bound or separation condition is required.

This estimate converts a spectral rank defect into an approximation by positive peaks in physical space. Positivity of the approximating momentum is essential here; an arbitrary operator truncation does not supply it. The exponent $4n$ cannot be reduced; see Proposition 4.6. To control the evolution of this approximation error, we introduce the coefficient norms and the weighted spectral defect below.

We write $[a]_+ = \max\{a, 0\}$ for $a \in \mathbb{R}$. The noise coefficients enter through

$$\begin{aligned} L_{j,k} &= \|\xi_k^{(j)}\|_\infty, \qquad 0 \leq j \leq 3, \\ \kappa_{1,k} &= 2(L_{1,k} + L_{3,k}), \\ \kappa_{2,k} &= 4(L_{1,k} + L_{3,k})(L_{0,k} + L_{1,k} + L_{2,k}). \end{aligned} \tag{1.3}$$

For $n \geq 1$, set

$$\mathcal{W}_n(m) = E(m)^{n-1} F_{n+1}(m),$$
$$\gamma_n = 2n\varepsilon^2 \sum_k ((4n-1)\kappa_{1,k}^2 + \kappa_{2,k}), \qquad V_n^* = 16n^2 \sum_k \kappa_{1,k}^2.$$

Theorem 1.2 (Multipeakon stability and estimates for exit times) *Fix an integer $n \geq 1$, a threshold $\eta > 0$ and $0 \leq T < \infty$. For every solution of Definition 2.1, $e^{-\gamma_n t}\mathcal{W}_n(m_t)$ is a nonnegative continuous supermartingale. Let*

$$\tau_{n,\eta}^* = \inf\{t : d_n(m_t) \geq \eta\}, \qquad b_{n,\eta}^* = \eta^{4n}/c_n.$$

Then

$$\mathbb{P}(\tau_{n,\eta}^* \leq T) \leq \min\{1, e^{\gamma_n T}\mathcal{W}_n(m_0)/b_{n,\eta}^*\}.$$

If $\mathcal{W}_n(m_0) > 0$ and $\varepsilon^2 V_n^* T > 0$, then

$$\mathbb{P}(\tau_{n,\eta}^* \leq T) \leq \exp\left\{-\frac{[\log(b_{n,\eta}^*/\mathcal{W}_n(m_0)) - \gamma_n T]_+^2}{2\varepsilon^2 V_n^* T}\right\}.$$

If $0 < \mathcal{W}_n(m_0) < b_{n,\eta}^*$ and $\gamma_n > 0$, then

$$\mathbb{E}\tau_{n,\eta}^* \geq \gamma_n^{-1} \log \frac{b_{n,\eta}^*}{\mathcal{W}_n(m_0)}.$$

If $\widetilde{m}_0$ has at most n atoms, $B = \|G * \widetilde{m}_0\|_{H^1}$, and $\|G * m_0 - G * \widetilde{m}_0\|_{H^1} \leq \delta$, then

$$\mathbb{P}(\tau_{n,\eta}^* \leq T) \leq \min\left\{1, c_n \frac{2^{n+1}}{n!} e^{\gamma_n T} \frac{(B+\delta)^{4n-2}\delta^2}{\eta^{4n}}\right\}. \tag{1.4}$$

For $n = 2$, the prefactor in (1.4) can be reduced to 576. There is no initial total-momentum restriction and no energy stopping time.

For fixed n, $\theta, \eta > 0$ and bounded B, (1.4) tends to zero with δ on $T = \theta\varepsilon^{-2}$, uniformly for $0 < \varepsilon \leq \varepsilon_*$. Corollary 5.2 extends the vanishing exit probability to intervals of length $a\varepsilon^{-2}\log(1/\delta)$ for a coefficient-dependent range of a. These estimates control proximity to the multipeakon set at each time. They do not determine whether the solution follows the trajectory issued from a specified nearby profile. The latter question requires a comparison of two solutions under the same Brownian driving.

Fix $n \geq 1$, deterministic $Q_1^0 < \cdots < Q_n^0$, and $P_i^0 > 0$. Let

$$v_t = \sum_{i=1}^n P_i(t) G(\cdot - Q_i(t))$$

be the multipeakon solution driven by the prescribed Brownian motions. Its positions remain distinct and its weights positive by Proposition 2.3.

Theorem 1.3 (Continuous dependence at multipeakon initial data) *Fix the reference solution above, $T > 0$, and the noise in (1.1). For any sequence of deterministic nonnegative finite initial momenta $m_0^{(j)}$ such that*

$$\delta_j = \|G * m_0^{(j)} - v_0\|_{H^1} \longrightarrow 0,$$

the solutions from Part I, driven by the same Brownian motions as v, satisfy, for every $1 \le p < \infty$,

$$\mathbb{E}\left[\sup_{t\le T}\|G * m_t^{(j)} - v_t\|_{H^1}^p\right] \longrightarrow 0.$$

There is no uniform bound on $m_0^{(j)}(\mathbb{R})$ and no total-momentum convergence assumption. If $\varepsilon = 0$ or all noise fields are spatially constant, then, for sufficiently small $\delta = \|G*m_0 - v_0\|_{H^1}$,

$$\left(\mathbb{E}\sup_{t\le T}\|u_t - v_t\|_{H^1}^2\right)^{1/2} \le C\delta^{1/2}.$$

For spatially varying noise, an explicit probability bound and a double-logarithmic modulus are given in Theorem 6.1. The constants depend on the fixed reference, T, ε_* and the noise norms, independently of the total momentum, number of atoms and spatial moments of the perturbation.

The convergence is equivalently in $L^p(\Omega; C([0,T]; H^1))$ for every finite p. The one-half exponent for spatially constant noise is sharp by Proposition 7.4. The following perturbation shows why initial H^1 convergence allows unbounded total momentum.

Remark Fix the reference momentum $n_0 = \sum_{i=1}^n P_i^0 \delta_{Q_i^0}$ and, for $L > 1$, set

$$\nu_L = L^{-3/4}\mathbf{1}_{[L,2L]}(x)\,\mathrm{d}x, \qquad m_0^{(L)} = n_0 + \nu_L.$$

Then $m_0^{(L)}(\mathbb{R}) = n_0(\mathbb{R}) + L^{1/4} \longrightarrow \infty$, whereas

$$\begin{aligned}\|G * m_0^{(L)} - G * n_0\|_{H^1}^2 &= \iint_{\mathbb{R}^2} G(x-y)\nu_L(\,\mathrm{d}x)\nu_L(\,\mathrm{d}y)\\ &= L^{-3/2}(L - 1 + e^{-L}) \longrightarrow 0.\end{aligned}$$

Indeed, $\int_L^{2L}\int_L^{2L} G(x-y)\,\mathrm{d}x\,\mathrm{d}y = L - 1 + e^{-L}$. This sequence satisfies the initial H^1 convergence hypothesis of Theorem 1.3, while convergence of the total masses, required for narrow convergence to n_0, fails.

The path comparison concerns velocities and admits non-atomic perturbations. To interpret a nearby finite multipeakon profile in terms of individual peaks, one needs a further geometric estimate: when the reference peaks are separated and their weights are bounded below, a small H^1 error determines the weights and positions locally.

Theorem 1.4 (Local estimates for weights and positions) *Fix*

$$U = \sum_{i=1}^n p_i G(\cdot - q_i), \qquad q_1 < \cdots < q_n, \qquad p_i \ge p_* > 0, \qquad q_{i+1} - q_i \ge \ell > 0.$$

For $n = 1$, the gap condition is vacuous. Let $V = \sum_{j=1}^s a_j G(\cdot - y_j)$ have at most n distinct ordered nodes with positive weights, allowing $s = 0$, and put $\delta = \|U - V\|_{H^1}$. For sufficiently small δ, one has $s = n$ and a unique local ordered matching, with

$$\left(\sum_{i=1}^n |a_i - p_i|^2\right)^{1/2} \le C\delta, \qquad \sum_{i=1}^n p_i^2|y_i - q_i| \le C\delta^2.$$

The threshold and constants depend only on p_* and ℓ; explicit values and the resulting gap estimates are given in Theorem 7.1. No upper weight bound, total-momentum bound or diameter bound is required.

The quadratic position estimate reflects the cusp of G: the squared H^1 distance between nearby translates is linear in their displacement. The theorem applies to the finite approximating profiles; the perturbed momentum in Theorem 1.3 may be non-atomic.

1.4 Main ideas of the proof

The set estimate combines a spatial approximation argument with relative stochastic bounds for its spectral defect. The first obstacle is positivity: a finite-rank truncation of the Green operator need not represent a nonnegative atomic momentum. We choose centers at eigenfunction zeros and project in H^1 onto the corresponding translates of G. The projection coefficients are nonnegative. Comparing the physical and spectral Dirichlet Green kernels then gives Theorem 1.1.

The second obstacle is to propagate a small spectral defect without assuming stochastic conservation of the CH integrals. The isospectral formulation of [12] requires $\xi_k^{(3)} = \xi_k'$, which restricts $C_b^3(\mathbb{R})$ coefficients to constants. We instead derive stochastic balance identities directly from the weak equation. Their tensor Itô formulas must remain valid on collision diagonals, where atomic measures contribute. A regularization argument preserves the cancellation of the deterministic CH drift, while relative Green-operator estimates give drift and diffusion bounds proportional to the exterior-power defect. These bounds yield the supermartingale and exit estimates of Theorem 1.2.

The path comparison uses a separate argument because a small set distance does not identify a reference trajectory. Its principal difficulty is the lack of total-momentum control illustrated by the diffuse perturbation above. We split the initial momentum into components localized near the reference peaks and a nonnegative remainder. An H^{-1} multiplier estimate controls the remainder's energy by the initial velocity error, even when its total momentum is large. Ordered characteristics transport the components, and self-interaction cancels in the remainder balance. The localized weight errors are normalized by a momentum threshold R, producing a growth exponent linear in R. This permits the choice $R \asymp \log(1/\delta)$. A logarithmic tail estimate removes the stop, and energy moments convert convergence in probability into convergence in every finite L^p path norm. For constant noise, a common random translation reduces the comparison to deterministic CH and gives the sharp one-half exponent.

The finite-profile parameter estimates follow from localized triangular and plateau tests. Separation and a lower weight bound first give one nearby node for each reference peak. The plateau tests control the weights; the triangular tests exploit the cusp of G to obtain the quadratic position estimate. This argument recovers parameters of finite approximating profiles and is independent of both stochastic comparisons.

1.5 Organization of the paper

Section 2 fixes the common solution framework from Part I. Section 3 develops the relative noise estimates for exterior powers. Section 4 turns the spectral defect into a positive multipeakon approximation, and Section 5 combines these estimates to prove set stability and exit bounds. Section 6 gives the independent localized comparison with a specified multipeakon; Section 7 interprets finite-profile errors through weights and positions. Section 8 discusses the relation between the results. Appendices A and B give the variants that use total momentum. Appendix C contains the longer Green-operator calculation used in Section 3.

2 Solution framework and results from Part I

2.1 Solutions with nonnegative momentum

We retain the notation $\mathcal{M}_+(\mathbb{R})$, G, u_m and M from the introduction. We set $G'(0) = 0$, the mean of the two one-sided limits of G' at the origin. The space $\mathcal{M}_+(\mathbb{R})$ carries the narrow topology: convergence is tested against all bounded continuous functions and includes convergence of total mass.

For $f \in C_c^\infty(\mathbb{R})$, set

$$
\begin{aligned}
B_k f &= \xi_k f' - \xi_k' f, \\
B_k^2 f &= \xi_k^2 f'' - \xi_k \xi_k' f' + \big((\xi_k')^2 - \xi_k \xi_k''\big) f, \\
F_f(m) &= \iint_{\mathbb{R}^2} K_f(x,y)\, m(\mathrm{d}x)\, m(\mathrm{d}y), \qquad (2.1) \\
K_f(x,y) &= \frac{1}{2} G(x-y)\big(f'(x) + f'(y)\big) - \frac{1}{2} G'(x-y)\big(f(x) - f(y)\big). \qquad (2.2)
\end{aligned}
$$

The last product in (2.2) is continuous across the diagonal. Hence (2.1) defines the nonlinear drift for measures with atoms without multiplying them by a discontinuous derivative.

Definition 2.1 A global weak martingale solution with nonnegative momentum and initial condition $m_0 \in \mathcal{M}_+(\mathbb{R})$ consists of a filtered probability space satisfying the usual conditions, an N-dimensional Brownian motion W, and an adapted process $m \in C([0,\infty); \mathcal{M}_+(\mathbb{R}))$ such that $m(0) = m_0$ and, for each $f \in C_c^\infty(\mathbb{R}) \cup \{1\}$,

$$
\begin{aligned}
\langle f, m_t \rangle = \langle f, m_0 \rangle &+ \int_0^t F_f(m_s)\, \mathrm{d}s + \frac{\varepsilon^2}{2} \sum_{k=1}^N \int_0^t \langle B_k^2 f, m_s \rangle\, \mathrm{d}s \\
&+ \varepsilon \sum_{k=1}^N \int_0^t \langle B_k f, m_s \rangle\, \mathrm{d}W_s^k. \qquad (2.3)
\end{aligned}
$$

The identity holds almost surely for all $t \geq 0$ for each fixed test function. We also require $\mathbb{E} \sup_{t \leq T} M(m_t)^p < \infty$ for every $T < \infty$ and $p \geq 2$.

For $u = G * m$, the energy introduced above satisfies

$$E(u) = \|u\|_{H^1}^2 = \iint_{\mathbb{R}^2} G(x-y)m(\,\mathrm{d}x)m(\,\mathrm{d}y), \qquad \mathcal{F}(u) = \int_{\mathbb{R}} (u^3 + uu_x^2)\,\mathrm{d}x.$$

We write $E(m) = E(G * m)$ when using momentum variables.

The unrestricted family $\mathcal{P}_n$ is defined in Section 1, and the family with prescribed total momentum in Appendix A.

Narrow convergence $m_j \to m$ in $\mathcal{M}_+(\mathbb{R})$ implies $G * m_j \to G * m$ in H^1. Indeed, expanding the squared H^1 difference gives three integrals of the bounded continuous kernel $G(x-y)$ against product measures, and each converges by narrow convergence and tightness. In particular, the velocities of all solutions in Definition 2.1 have continuous H^1 paths.

2.2 Global well-posedness

A strong solution is understood in the probabilistic sense: it is constructed on the prescribed probability space with the given Brownian motion. The momentum may have atoms, and the velocity need not be a classical solution. We use the solution class in Definition 2.1.

Theorem 2.2 (Global solution theory from Part I) *For every deterministic $m_0 \in \mathcal{M}_+(\mathbb{R})$, there is a global solution in the sense of Definition 2.1, adapted to the completed Brownian filtration on any prescribed Brownian probability space. Any two solutions in that definition on the same space with the same m_0 and W are indistinguishable, and the solution is unique in law. It has the representation*

$$m_t = (q_t)_{\#}(p_t\, m_0), \qquad q_0(x) = x, \quad p_0(x) = 1.$$

The map q_t is strictly increasing on all of $\mathbb{R}$, and for every $T < \infty$, almost surely,

$$\sup_{t\le T}\|q_t - \mathrm{id}\|_\infty < \infty, \qquad \Big\| \sup_{t\le T}\big(p_t + p_t^{-1}\big)\Big\|_{L^\infty(m_0)} < \infty.$$

For m_0-almost every label, the characteristic equations are

$$\begin{aligned} \mathrm{d}q(x) &= u(q(x))\,\mathrm{d}t + \varepsilon \sum_k \xi_k(q(x)) \circ \mathrm{d}W^k, \\ \mathrm{d}p(x) &= -p(x)\bar{u}_x(q(x))\,\mathrm{d}t - \varepsilon p(x) \sum_k \xi_k'(q(x)) \circ \mathrm{d}W^k. \end{aligned} \tag{2.4}$$

Here $\bar{u}_x = G' * m$ is defined using $G'(0) = 0$.

Theorem 1.1 of Part I [22] proves this result, including the characteristic representation for every solution of Definition 2.1. We use the position flow defined there for every initial label; the first equation in (2.4) holds almost surely for each fixed label, including labels not charged by m_0. The weight equation is required only m_0-almost everywhere. Corollary 4.2 of Part I shows that narrow convergence of deterministic initial momenta implies convergence in probability in the narrow path topology on each finite interval and, after convolution with G, in $C([0,T];H^1)$. In particular, positive

particle approximations converge on the prescribed Brownian space. Narrow convergence includes convergence of total momentum. In Section 6, continuous dependence is established using only initial H^1 velocity convergence.

Proposition 2.3 (Positive particle solutions) *If $m_0 = \sum_{i=1}^{s} p_i^0 \delta_{q_i^0}$ has positive weights and strictly ordered positions, the solution remains of this form and satisfies*

$$\begin{aligned} \mathrm{d}q_i &= u(q_i)\,\mathrm{d}t + \varepsilon \sum_k \xi_k(q_i) \circ \mathrm{d}W^k, \\ \mathrm{d}p_i &= -p_i \bar{u}_x(q_i)\,\mathrm{d}t - \varepsilon p_i \sum_k \xi_k'(q_i) \circ \mathrm{d}W^k. \end{aligned}$$

The positions remain strictly ordered and the weights positive on every finite time interval. For each $p \geq 2$ and $T < \infty$,

$$\mathbb{E} \sup_{t \leq T} M(m_t)^p \leq C_{p,\xi,\varepsilon,T} M(m_0)^p, \tag{2.5}$$

with a constant independent of the number and positions of the particles.

Proof The representation in Theorem 2.2 gives the particle equations, positivity and strict ordering. Testing (2.3) with $f = 1$ gives, for any solution in that definition,

$$\mathrm{d}M = -\varepsilon \sum_k \langle \xi_k', m \rangle \,\mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k \langle (\xi_k')^2 - \xi_k \xi_k'', m \rangle \,\mathrm{d}t.$$

The absolute value of the drift is bounded by a constant times $\varepsilon^2 M$, and the diffusion coefficients are bounded by constants times εM. Localization, the Burkholder–Davis–Gundy inequality and Gronwall's inequality prove (2.5). □

The energy balance follows from Lemma 3.2 with $j = 2$, since $p_2 = 2E$. The tensor formula and regularization below extend the particle identities to the full weak solution class.

3 Spectral estimates

Exterior powers measure the defect from finite rank; relative derivative bounds control its stochastic evolution independently of total momentum. Section 4 converts these bounds into positive multipeakon approximation.

3.1 Spectral and spatial quantities

Let $\mathcal{H}$ be $H^1(\mathbb{R})$ equipped with $\|f\|_{\mathcal{H}}^2 = \int_{\mathbb{R}} (f_x^2 + f^2/4)\,\mathrm{d}x$, whose reproducing kernel is $k(x,y) = e^{-|x-y|/2}$. The evaluation map $j : \mathcal{H} \to L^2(m)$ is Hilbert–Schmidt, with $\|j\|_{\mathrm{HS}}^2 = M$. Hence

$$(T_m f)(x) = \int_{\mathbb{R}} k(x,y) f(y) m(\,\mathrm{d}y) = j j^* f$$

is the positive trace-class operator used in the introduction. With its eigenvalues λ_i as defined there, put

$$p_j(m) = \operatorname{tr}(T_m^j), \quad e_0 = 1, \quad e_r(m) = \sum_{i_1<\cdots<i_r} \lambda_{i_1} \cdots \lambda_{i_r}.$$

These quantities are finite. The energy identity $\langle G(\cdot - x), G(\cdot - y)\rangle_{H^1} = G(x-y)$ gives

$$p_1 = M, \quad p_2 = 2E, \quad p_3 = 6\mathcal{F}, \quad \mathcal{F}(u) = \int_{\mathbb{R}} (u^3 + uu_x^2)\,\mathrm{d}x, \qquad D = \tfrac{1}{2}M^2 - E = e_2.$$

For the cubic identity, we use the representation $6\mathcal{F} = \int_{\mathbb{R}^3} e^{-(\max(x,y,z)-\min(x,y,z))}\,\mathrm{d}m^{\otimes 3}$; it follows by splitting into the six ordering regions and extending to repeated coordinates by continuity.

On the fixed space $\mathcal{H}$, put $k_x = k(\cdot, x)$ and

$$\mathcal{C}_m = \int_{\mathbb{R}} k_x \otimes k_x \, m(\mathrm{d}x).$$

This is a Bochner integral in the trace class: $\|k_x\|_{\mathcal{H}} = 1$, and $\|k_x - k_y\|_{\mathcal{H}}^2 = 2(1 - e^{-|x-y|/2})$. Thus $\mathcal{C}_m = j^*j$ and its nonzero spectrum agrees with that of $T_m = jj^*$. Moreover,

$$\overline{\operatorname{ran} \mathcal{C}_m} = \overline{\operatorname{span}\{k_x : x \in \operatorname{supp} m\}}.$$

Indeed, its orthogonal complement consists of the continuous functions $f \in \mathcal{H}$ with $\int_{\mathbb{R}} |f|^2\,\mathrm{d}m = 0$, which are exactly those vanishing on $\operatorname{supp} m$. The functions $k_{x_1}, \ldots, k_{x_r}$ at distinct points are linearly independent: applying $-\partial_x^2 + 1/4$ to a vanishing linear combination gives a vanishing linear combination of the distinct Dirac measures. Consequently $\mathcal{C}_m$ has finite rank s exactly when m has s support points. For finite nonnegative measures $m, \widetilde{m}$ one has

$$\|\mathcal{C}_m - \mathcal{C}_{\widetilde{m}}\|_{\mathrm{HS}}^2 = \iint_{\mathbb{R}^2} e^{-|x-y|}\,\mathrm{d}(m-\widetilde{m})(x)\,\mathrm{d}(m-\widetilde{m})(y) = 2\|G*m - G*\widetilde{m}\|_{H^1}^2. \tag{3.1}$$

To obtain this identity, we take Hilbert–Schmidt inner products of the rank-one operators and use the integrability of their trace norms.

3.2 Noise deformation of the Green operator

The relative derivative estimate of Part I [22, Lemma 5.5] controls all exterior powers independently of total momentum and operator rank. We state its fixed-space form below and give the full proof in Appendix C. Fix one $\xi \in C_b^3$ and consider its deterministic flow $\partial_s \phi_s = \xi(\phi_s)$, $\phi_0 = \mathrm{id}$, with Jacobian $J_s = \partial_x \phi_s$. The coadjoint action on momentum is $m_s = (\phi_s)_{\#}(J_s^{-1} m)$. The operator T_{m_s} is unitarily equivalent to the following operator on the fixed space $L^2(m)$:

$$T_s(x,y) = \frac{\exp(-|\phi_s(x) - \phi_s(y)|/2)}{\sqrt{J_s(x) J_s(y)}}. \tag{3.2}$$

The unitary map sends $g \in L^2(m_s)$ to $g(\phi_s(\cdot))/\sqrt{J_s(\cdot)}$. Order preservation fixes the sign of each spatial difference. The kernels in (3.2), including their diagonal values, are twice continuously differentiable in s, with derivatives bounded locally in s. Since m has finite mass, dominated convergence gives C^2 regularity in Hilbert–Schmidt norm.

Lemma 3.1 (Relative bounds for derivatives of the Green operator) *At $s = 0$, there are bounded self-adjoint operators K_1, K_2 on $L^2(m)$ such that*

$$T_0' = T_m^{1/2} K_1 T_m^{1/2}, \qquad T_0'' = T_m^{1/2} K_2 T_m^{1/2}, \qquad \|K_1\| \le \kappa_1, \quad \|K_2\| \le \kappa_2, \tag{3.3}$$

with κ_1, κ_2 given by (1.3) *for ξ.*

For the variation identity behind the estimate, fix real $f \in L^2(m)$ and set

$$n_s = (\phi_s)_\#(J_s^{-1/2} f\, m), \qquad v_s = (-\partial_x^2 + 1/4)^{-1} n_s, \qquad Q_s = \langle f, T_s f\rangle = \|v_s\|_{\mathcal{H}}^2.$$

Then

$$Q_s' = \frac{1}{2}\int_{\mathbb{R}} (\xi''' - \xi') v_s^2 \,\mathrm{d}x.$$

The L^2 derivative bound for v_s gives the second variation without differentiating ξ'''. The resulting quadratic-form bounds yield the factorizations in (3.3), including on the kernel of T_m. Appendix C justifies these steps for finite signed measures.

3.3 Tensor Itô formula

For the solution class in Section 2, the weak product argument gives the following tensor identity for every smooth kernel h with bounded derivatives through order two:

$$\begin{aligned} \mathrm{d}\int_{\mathbb{R}^j} h \,\mathrm{d}m^{\otimes j} = \sum_{i=1}^{j} \int_{\mathbb{R}^j} [u(x_i)\partial_i h - \bar{u}_x(x_i) h]\,\mathrm{d}m^{\otimes j}\,\mathrm{d}t \\ + \varepsilon \sum_k \int_{\mathbb{R}^j} \mathcal{T}_{\xi_k}^{(j)} h \,\mathrm{d}m^{\otimes j}\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k \int_{\mathbb{R}^j} (\mathcal{T}_{\xi_k}^{(j)})^2 h \,\mathrm{d}m^{\otimes j}\,\mathrm{d}t, \end{aligned} \tag{3.4}$$

where $\mathcal{T}_\xi^{(j)} = \sum_{i=1}^j (\xi(x_i)\partial_i - \xi'(x_i))$ and diagonal terms use $G'(0) = 0$. For products of compactly supported smooth functions, the formula follows from the ordinary Itô product formula. Finite tensor sums are dense in the uniform C^2 norm on compact boxes. Separate spatial cutoffs remove the compact-support restriction. The total-momentum moments control the drift and stochastic integrals, while compactness of each narrowly continuous path gives uniform spatial tightness. The collision-kernel regularization required here is proved in Section 3.4.

Primes on functionals denote differentiation along the deterministic flow used in Lemma 3.1; a subscript k selects the coefficient ξ_k.

3.4 Stochastic balances for spectral traces

Recall that, for integers $j \geq 1$,

$$p_j(m) = \operatorname{tr}(T_m^j), \qquad C_j(\boldsymbol{x}) = \exp\left(-\frac{1}{2}\sum_{i=1}^{j}|x_i - x_{i+1}|\right), \quad x_{j+1} = x_1.$$

For $j = 1$ this means $C_1 = 1$ and $p_1 = M$. Kernel multiplication gives $p_j = \int_{\mathbb{R}^j} C_j \, \mathrm{d}m^{\otimes j}$. Let S_j denote the average of C_j over permutations of its arguments, so the same integral is obtained with S_j.

Lemma 3.2 (Stochastic balances for all traces) *For every fixed $j \geq 1$, on the given Brownian space,*

$$\mathrm{d}p_j = \varepsilon \sum_k p'_{j,k}\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k p''_{j,k}\,\mathrm{d}t, \qquad p^{(\ell)}_{j,k} = \int_{\mathbb{R}^j} (\mathcal{T}^{(j)}_{\xi_k})^\ell S_j \, \mathrm{d}m^{\otimes j}, \quad \ell = 1, 2.$$

In particular, the deterministic CH drift is zero. All these identities can be imposed simultaneously on a single event of probability one.

Proof We first identify the deterministic cancellation, including on collision sets. We consider finitely many positive particles with distinct positions. Set

$$L_{ab} = \sqrt{p_a p_b}\, e^{-|q_a - q_b|/2}, \qquad \mathbb{B}_{ab} = \tfrac{1}{4}\operatorname{sgn}(q_a - q_b) L_{ab}.$$

Direct substitution of the deterministic particle equations gives $\dot{L} = [\mathbb{B}, L]$. To verify the normalization, put $s_{ab} = \operatorname{sgn}(q_a - q_b)$ and $d_{ab} = |q_a - q_b|$. For each matrix entry, the required identity is

$$(s_{ac} + s_{bc})e^{-(d_{ac}+d_{bc}-d_{ab})/2} = (s_{ac} - s_{ab})e^{-d_{ac}} + (s_{bc} + s_{ab})e^{-d_{bc}}.$$

The identity follows by distinguishing whether q_c lies inside or outside the interval with endpoints q_a, q_b; it also holds for repeated indices. Thus $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr} L^j = 0$.

We next express the particle identity in terms of continuous $(j+1)$-point kernels. The transport part of the deterministic tensor drift, after inserting an additional variable y, is the permutation average of

$$-\tfrac{1}{2} C_j(\boldsymbol{x}) \sum_i \operatorname{sgn}(x_i - x_{i+1})[G(x_i - y) - G(x_{i+1} - y)].$$

The kernel is bounded and continuous, even at collisions. Write $S_{-\ell} = S_j(r_1, \ldots, \widehat{r}_\ell, \ldots, r_{j+1})$. The fully symmetrized reaction part is

$$-\frac{1}{j+1}\sum_{\ell<i} G'(r_i - r_\ell)(S_{-\ell} - S_{-i}).$$

The difference in parentheses is Lipschitz and vanishes when $r_i = r_\ell$. Thus the reaction kernel is also continuous under the convention $G'(0) = 0$. Let Q_j be the resulting symmetric full drift kernel. For any distinct $r_1, \ldots, r_{j+1}$, apply the particle identity to $m = \sum_a w_a \delta_{r_a}$ for arbitrary positive w_a. The coefficient of $w_1 \cdots w_{j+1}$ in the zero polynomial $\int_{\mathbb{R}^{j+1}} Q_j \, \mathrm{d}m^{\otimes(j+1)}$ is $(j+1)!\, Q_j(r_1, \ldots, r_{j+1})$. Hence $Q_j = 0$ off the collision set, and continuity gives $Q_j = 0$

everywhere. Thus the deterministic cancellation holds for arbitrary finite positive measures, including measures with non-atomic support.

We next identify the continuous noise kernels and justify their use in the weak equation for measures with atoms. Put

$$a_\xi = \tfrac{1}{2}\sum_i \operatorname{sgn}(x_i - x_{i+1})[\xi(x_i) - \xi(x_{i+1})], \quad b_\xi = \sum_i \xi'(x_i),$$

and define w_ξ by replacing ξ in the difference in a_ξ by $\xi\xi'$, and $d_\xi = \sum_i \xi(x_i)\xi''(x_i)$. Then

$$\mathcal{T}_\xi^{(j)} C_j = -(a_\xi + b_\xi)C_j, \qquad (\mathcal{T}_\xi^{(j)})^2 C_j = [(a_\xi + b_\xi)^2 - w_\xi - d_\xi]C_j.$$

Both kernels are bounded and continuous. Their permutation averages give the corresponding formulas for S_j.

Choose a nonnegative smooth mollifier supported in $[-1,1]^j$ and invariant under permutations, and write $S_{j,\rho}$ for its convolution with S_j at scale $\rho \in (0,1]$. Then $\|S_{j,\rho} - S_j\|_\infty \le C_j\rho$. We use the following estimates to justify regularization when m has atoms. Away from the collision hyperplanes, the Hessian of C_j is the sum of bounded regular terms. Its singular part is a finite sum of bounded surface densities on $x_i = x_\ell$, each multiplied by $(e_i - e_\ell)\otimes(e_i - e_\ell)$. After convolution, each such density is bounded by C_j/ρ and is supported where $|x_i - x_\ell| \le 2\rho$. Contraction with the noise vector therefore bounds its contribution by $C_{j,\xi}\rho$, since $|\xi(x_i) - \xi(x_\ell)| \le L_1|x_i - x_\ell|$.

For the regular terms, contracting the gradient of C_j with the noise vector gives $-a_\xi C_j$, while contraction of the regular Hessian gives $a_\xi^2 C_j$. In a convolution integral, replacing a coefficient at $\boldsymbol{x}$ by its value at the integration point changes it by $O(\rho)$. The remaining first-order contraction uses the same expression with ξ replaced by $\xi\xi'$. These expressions are Lipschitz under the stated assumptions, including across collisions, since each sign is multiplied by the corresponding difference. The bounded derivatives of ξ then give

$$\left\| (\mathcal{T}_\xi^{(j)})^\ell S_{j,\rho} - (\mathcal{T}_\xi^{(j)})^\ell S_j \right\|_\infty \le C_{j,\xi}\rho, \qquad \ell = 1,2,$$

where the limiting kernels are the continuous formulas displayed above.

For the deterministic transport term, write every derivative of C_j as a sum of signed edge terms and pair the endpoints of each edge. If mollification changes the sign of $x_i - x_\ell$, then $|x_i - x_\ell| \le 2\rho$, so $|u(x_i) - u(x_\ell)| \le M\rho$. If the sign does not change, the Lipschitz bound for C_j gives the same error bound. The reaction term changes by at most $C_j\rho M$ because $\|\bar{u}_x\|_\infty \le M/2$. After integration against $m^{\otimes j}$, the deterministic drift therefore differs from the integral of Q_j by at most $C_j\rho M^{j+1}$, and that integral is zero. We apply (3.4) to $S_{j,\rho}$ and pass to the limit using the preceding bounds and the moments of $\sup_{t\le T} M_t$. The drift integrals converge in L^1 and the stochastic integrals in L^2, uniformly on bounded time intervals in probability. The functionals themselves converge uniformly on each such interval. Hence the identity holds on the original probability space. Using continuity and taking a countable intersection over j, we obtain the asserted common event. □

3.5 Estimates for exterior powers

The exterior-power functionals provide defects from finite rank. Their relative drift and noise bounds vanish with the defects themselves, so the stochastic estimates remain valid as eigenvalues approach zero. We also bound their initial values by the squared H^1 perturbation. Section 4 converts this spectral control into approximation by positive multipeakons. For $r \ge 2$ define

$$F_r(m) = e_r(\lambda_1(T_m)^2, \lambda_2(T_m)^2, \ldots) = \|\textstyle\bigwedge^r T_m\|_{\mathrm{HS}}^2.$$

It vanishes precisely for measures supported on at most $r-1$ points. Newton's identities express F_r through $p_2, p_4, \dots, p_{2r}$, so its exact stochastic balance for every weak solution follows from Lemma 3.2.

Proposition 3.3 (Estimates for exterior powers) *For the constants $\kappa_{1,k}, \kappa_{2,k}$ in* (1.3)*,*

$$\begin{aligned} |F'_{r,k}| &\le 2r\kappa_{1,k}F_r, \\ |F''_{r,k}| &\le 2r\big((2r-1)\kappa_{1,k}^2 + \kappa_{2,k}\big)F_r. \end{aligned} \tag{3.5}$$

Consequently $e^{-\Lambda_r t}F_r(m_t)$ is a nonnegative supermartingale, where

$$\Lambda_r = \varepsilon^2 r \sum_k ((2r-1)\kappa_{1,k}^2 + \kappa_{2,k}).$$

If $\widetilde{m}$ has at most $r-1$ atoms and $\|G*m - G*\widetilde{m}\|_{H^1} \le \delta$, then

$$F_r(m) \le \frac{2^r}{(r-1)!}E(m)^{r-1}\delta^2 \le \frac{2^r}{(r-1)!}\big(\|G*\widetilde{m}\|_{H^1} + \delta\big)^{2r-2}\delta^2. \tag{3.6}$$

When $r = 3$, the coefficient $2^3/2! = 4$ can be replaced by 2. These estimates require no bound on total momentum.

Proof At a fixed parameter value along a deterministic noise flow, write $T' = T^{1/2}K_1T^{1/2}$ and $T'' = T^{1/2}K_2T^{1/2}$ as in (3.3). For $S = \bigwedge^r T$, differentiation of the finite tensor product gives relative factorizations $S' = S^{1/2}AS^{1/2}$ and $S'' = S^{1/2}BS^{1/2}$ with

$$\|A\| \le r\kappa_1, \qquad \|B\| \le r\kappa_2 + r(r-1)\kappa_1^2.$$

To see this, insert K_1 successively in each tensor factor for the first derivative. For the second derivative, insert K_2 in one factor or K_1 in two distinct factors. Then restrict the symmetric sum to the antisymmetric tensor space. The Hilbert–Schmidt differentiability of the tensor product justifies this calculation without an inverse of T. The Schatten norm inequality $\|S^{1/2}AS^{1/2}\|_{\mathrm{HS}} \le \|A\|\,\|S\|_{\mathrm{HS}}$ gives

$$\begin{aligned} |F'_r| &= 2|\operatorname{tr}(SS')| \le 2\|A\|\,F_r, \\ |F''_r| &\le 2\|S'\|_{\mathrm{HS}}^2 + 2|\operatorname{tr}(SS'')| \le 2(\|A\|^2 + \|B\|)F_r. \end{aligned}$$

These bounds prove (3.5), including at operators of rank less than r, without dividing by an eigenvalue.

For the initial estimate, set $n = r-1$, $C = \mathcal{C}_m$, $\widetilde{C} = \mathcal{C}_{\widetilde{m}}$, $x_i = \lambda_i(C)^2$, $A = \sum_i x_i = 2E(m)$ and $t = \sum_{i>n} x_i$. Let P be the orthogonal projection onto the range of $\widetilde{C}$, which has dimension at most n. Then

$$\|C - \widetilde{C}\|_{\mathrm{HS}}^2 \ge \|(I-P)C\|_{\mathrm{HS}}^2 = A - \operatorname{tr}(PC^2) \ge t.$$

For the last inequality, in an eigenbasis v_i of C the coefficients $\|Pv_i\|^2$ lie in $[0,1]$ and sum to at most n, so $\operatorname{tr}(PC^2) \le \sum_{i\le n} x_i$. Every set of $n+1$ indices contains an index greater than n. Therefore

$$F_{n+1} = e_{n+1}(x) \le \sum_{i>n} x_i e_n(x_1, \dots, \widehat{x_i}, \dots) \le t\, e_n(x) \le \frac{A^n t}{n!}.$$

Tonelli's theorem justifies these sums, and the final bound follows by expanding A^n. The isometry (3.1) now proves (3.6). For $n = 2$, write $h = x_1 + x_2$ and $t = \sum_{i>2} x_i$. The sharper calculation

$$\begin{aligned} F_3 &= x_1x_2t + h\,e_2(x_{>2}) + e_3(x_{>2}) \\ &\le \frac{h^2t}{4} + \frac{ht^2}{2} + \frac{t^3}{6} = t\left(\frac{A^2}{4} - \frac{t^2}{12}\right) \le \frac{A^2t}{4} \end{aligned}$$

gives $F_3 \le 2E^2\delta^2$. The exact scalar balance and the coefficient bounds give the supermartingale property. □

The estimates also control approximation by finite-rank operators. For $n \ge 1$, $\eta > 0$ and $0 \le T < \infty$, put $s_n(m)^2 = \sum_{i>n} \lambda_i(T_m)^2$ and $r = n + 1$. Then

$$s_n^{2r} \le r!F_r, \qquad \mathbb{P}\left(\sup_{t\le T} s_n(m_t) \ge \eta\right) \le \min\{1, r!e^{\Lambda_r T}F_r(m_0)/\eta^{2r}\}.$$

For the first inequality, expand $r!e_r$ as a sum of products of the nonnegative numbers λ_i^2 over ordered distinct indices. After excluding any $j \le n$ indices, the remaining sum is at least s_n^2. Applying this observation successively gives the bound. The second inequality follows from the maximal inequality. This estimate concerns approximation by arbitrary rank-n operators. The next section uses the one-dimensional Green kernel to construct positive atomic velocities from the spectral information. The spectral distance s_n has continuous paths: by (3.1) the fixed-space operator has continuous Hilbert–Schmidt paths, and distance to the set of operators of rank at most n is Lipschitz in that norm.

4 Approximation by positive multipeakons

For finite atomic momenta, we choose centers at zeros of a spectral eigenfunction and project the velocity onto the corresponding Green translates. This projection has nonnegative coefficients. Comparing the Dirichlet kernels of $-\partial_x^2 + 1/4$ and $-\partial_x^2 + 1$ transfers the spectral bound to the physical H^1 error. For general momenta, we pass this bound to the limit using continuity of the spatial and spectral distances, without requiring convergence of the selected centers. Write

$$s_j(m)^2 = \sum_{\ell>j} \lambda_\ell(T_m)^2.$$

Total momentum is not prescribed in the definition of d_n.

4.1 Orthogonal projection and comparison of Green kernels

For $\alpha > 0$, let

$$g_\alpha(x, y) = \frac{e^{-\alpha|x-y|}}{2\alpha}.$$

This is the Green kernel of $-\partial_x^2 + \alpha^2$ on the line. Thus $g_1(x, y) = G(x - y)$ and $g_{1/2}(x, y) = k(x, y)$. Given distinct centers $Q = \{q_1 < \cdots < q_j\}$, let $g_{\alpha,Q}$ be the

Dirichlet Green kernel on the components of $\mathbb{R} \setminus Q$, extended by zero for points in different components or in Q. It is also the reproducing kernel of the closed subspace

$$\{f \in H^1(\mathbb{R}) : f(q_a) = 0,\ 1 \le a \le j\}$$

for the norm $\int_{\mathbb{R}} (f_x^2 + \alpha^2 f^2)\,\mathrm{d}x$.

Lemma 4.1 (Positivity of the projected momentum) *For $m \neq 0$, the H^1 orthogonal projection $\Pi_Q u_m$ onto $\operatorname{span}\{G(\cdot - q_a) : 1 \le a \le j\}$ is a positive multipeakon with at most j peaks, after removing zero coefficients. The zero measure is treated by $d_j(0) = 0$. Moreover,*

$$d_j(m)^2 \le \|u_m - \Pi_Q u_m\|_{H^1}^2 = \iint_{\mathbb{R}^2} g_{1,Q}(x,y) m(\mathrm{d}x) m(\mathrm{d}y). \tag{4.1}$$

Proof It suffices to project one Green translate. If $q_a < x < q_{a+1}$, the projection of $G(\cdot - x)$ has just the two coefficients

$$\frac{\sinh(q_{a+1} - x)}{\sinh(q_{a+1} - q_a)}, \qquad \frac{\sinh(x - q_a)}{\sinh(q_{a+1} - q_a)}$$

at q_a, q_{a+1}. If $x < q_1$ or $x > q_j$, its only coefficient is $e^{-|x - q_{\mathrm{nearest}}|}$ at the nearest center; at a center the projection is the same Green translate. These formulas follow by matching the values at the centers and using the reproducing identity: evaluation is the H^1 inner product with the corresponding Green translate. The coefficients are nonnegative and sum to at most one, so they are integrable against the finite nonnegative measure m. Integrating these projections gives the velocity of a positive atomic momentum measure. The residual kernel is $g_{1,Q}$ by the reproducing-kernel projection formula, which gives (4.1). □

Lemma 4.2 (Green-kernel comparison) *For every nonempty finite Q and every $x, y \in \mathbb{R}$,*

$$0 \le g_{1,Q}(x,y) \le 2\, k(x,y) g_{1/2,Q}(x,y).$$

When Q consists of one point, the factor 2 can be replaced by 1.

Proof Both Dirichlet kernels vanish when their arguments lie in different components, so consider $x \le y$ in the same component. On a bounded component (a,b), put $A = x - a$, $B = b - y$, $C = y - x$ and $L = A + B + C$. Then

$$g_{1,Q}(x,y) = \frac{\sinh A \sinh B}{\sinh L}, \qquad g_{1/2,Q}(x,y) = \frac{2\sinh(A/2)\sinh(B/2)}{\sinh(L/2)}.$$

For $A, B > 0$ their ratio satisfies

$$\frac{g_{1,Q}(x,y)}{k(x,y) g_{1/2,Q}(x,y)} = \frac{e^{C/2}\cosh(A/2)\cosh(B/2)}{\cosh(L/2)} \le 2.$$

We use $\cosh t \le e^t$ and $\cosh(L/2) \ge e^{L/2}/2$. The endpoint cases follow by continuity. On a half-line, if A is the distance of the nearer point from the finite endpoint, the same ratio is $(1 + e^{-A})/2 \le 1$. For a single center all components are half-lines. □

4.2 Eigenfunction zeros and Dirichlet Green operators

Define the operator $T_{m,Q}$ on $L^2(m)$ by the kernel $g_{1/2,Q}(x,y)$. Let $\iota_m : \mathcal{H} \to L^2(m)$ be evaluation, $\iota_m f = f$, with $\mathcal{H}$ as in Section 3. If P_Q is orthogonal projection in $\mathcal{H}$ onto the span of $k(\cdot, q_a)$, then

$$T_{m,Q} = \iota_m (I - P_Q) \iota_m^*, \qquad 0 \le T_{m,Q} \le T_m.$$

Lemma 4.3 (Bounds for the Dirichlet Green operator) *Suppose m has $L > j$ distinct support points with positive weights. There are j distinct centers Q such that*

$$\|T_{m,Q}\|_{\mathrm{op}} \le \lambda_{j+1}(T_m), \qquad \|T_{m,Q}\|_{\mathrm{HS}}^2 \le (j+1) s_j(m)^2. \tag{4.2}$$

Proof Write $m = \sum_{a=1}^{L} w_a \delta_{x_a}$, $x_1 < \cdots < x_L$, and set $P = \mathrm{diag}(w_a)$, $K_{ab} = e^{-|x_a - x_b|/2}$. The matrix representing T_m in an orthonormal basis of $L^2(m)$ is $S = P^{1/2} K P^{1/2}$. Let $\rho_a = e^{-(x_{a+1} - x_a)/2} \in (0,1)$. Direct multiplication shows that K^{-1} is tridiagonal with

$$\begin{aligned}
(K^{-1})_{a,a+1} &= -\frac{\rho_a}{1-\rho_a^2}, \\
(K^{-1})_{11} = \frac{1}{1-\rho_1^2}, &\qquad (K^{-1})_{LL} = \frac{1}{1-\rho_{L-1}^2}, \\
(K^{-1})_{aa} &= \frac{1}{1-\rho_{a-1}^2} + \frac{\rho_a^2}{1-\rho_a^2}, \qquad 1 < a < L.
\end{aligned}$$

Thus S^{-1} is an irreducible symmetric Jacobi matrix with strictly negative adjacent entries.

By the finite Jacobi oscillation theorem, the eigenvector for the kth increasing eigenvalue has $k-1$ nodes; zero components count as nodes, with opposite signs on either side. See [21, Theorem 4.7, equation (4.38)]. The finite-dimensional count can also be obtained directly from the leading-principal determinants: for diagonal entries a_ℓ and adjacent entries $-b_\ell$, $b_\ell > 0$, they obey

$$\Delta_0 = 1, \quad \Delta_1 = a_1 - \mu, \quad \Delta_\ell = (a_\ell - \mu)\Delta_{\ell-1} - b_{\ell-1}^2 \Delta_{\ell-2}.$$

The eigenvector components are proportional to $\Delta_{\ell-1}/(b_1 \cdots b_{\ell-1})$. Gaussian elimination equates the sign variations of this Sturm chain with the negative inertia below μ. At an interior zero the adjacent determinants have opposite signs, so deleting zeros preserves the count. At $\mu = \mu_k$ the terminal zero $\Delta_L(\mu_k) = 0$ is excluded; the interior-node count is the limit from $\mu < \mu_k$ and equals $k-1$. Irreducibility also gives simple eigenvalues, nonzero end components, and no consecutive zero components.

Take $k = j+1$, and let v be an eigenvector of S for its kth decreasing eigenvalue λ_k. Extend it to the line by

$$\psi(x) = \sum_{a=1}^{L} \sqrt{w_a}\, v_a k(x, x_a), \qquad \psi(x_a) = \lambda_k v_a / \sqrt{w_a}.$$

Then $-\psi'' + \psi/4 = \lambda_k^{-1} \psi\, m$ in distributions. Between neighboring support points ψ is the hyperbolic interpolant of its endpoint values. It has precisely one zero between opposite nonzero endpoint signs and none between equal signs. A zero at an atom has opposite neighboring signs; the derivative jump there is $-\lambda_k^{-1} w_a \psi(x_a) = 0$, so the zero is a simple crossing. There are no zeros on the outer exponential tails. Consequently ψ has exactly j distinct zeros on the line. Take these zeros as Q.

On each component I of $\mathbb{R}\setminus Q$, ψ has a constant strict sign. Its restriction belongs to $H^1_0(I)$, with the usual decay condition at an infinite endpoint. Restricting the distributional eigenvalue equation to I and applying the Dirichlet inverse of $-\partial_x^2 + 1/4$ gives the integral equation

$$\int_I g_{1/2,Q}(x,y)\psi(y)m(\,\mathrm{d}y) = \lambda_k\psi(x), \qquad x \in I.$$

The vector of its nonzero values at the atoms has entries of the same sign and is an eigenvector of the strictly positive matrix associated with this component. By the Perron–Frobenius theorem, λ_k is therefore the largest eigenvalue of that block. Atoms in Q give zero rows and columns, and components with zero momentum measure contribute zero operators. This proves $\|T_{m,Q}\|_{\mathrm{op}} \le \lambda_k$.

Let μ_ℓ be the decreasing eigenvalues of $T_{m,Q}$. The positive-operator order $T_{m,Q} \le T_m$ and the min–max principle give $\mu_\ell \le \lambda_\ell$. Hence

$$\sum_\ell \mu_\ell^2 \le j\lambda_{j+1}^2 + \sum_{\ell>j}\lambda_\ell^2 \le (j+1)s_j^2.$$

This yields (4.2). □

Proposition 4.4 (Approximation error from a spectral tail) *For every finite nonnegative measure,*

$$d_1(m)^4 \le 4E(m)s_1(m)^2, \tag{4.3}$$

$$d_j(m)^4 \le 8(j+1)E(m)s_j(m)^2, \qquad j \ge 2. \tag{4.4}$$

Proof For a finite atomic measure with more than j support points, choose the nodal centers given by Lemma 4.3. Lemmas 4.1 and 4.2, followed by Cauchy–Schwarz in $L^2(m\otimes m)$, give

$$d_j^2 \le 2\iint_{\mathbb{R}^2} k(x,y)g_{1/2,Q}(x,y)\,\mathrm{d}m(x)\,\mathrm{d}m(y) \le 2\sqrt{2E}\,\|T_{m,Q}\|_{\mathrm{HS}} \le 2\sqrt{2(j+1)E}\,s_j.$$

For $j = 1$, the Green-kernel comparison has factor one, yielding the first bound. Measures with at most j atoms have $d_j = 0$.

For arbitrary finite nonnegative m, choose finite positive atomic $m_\ell \to m$ narrowly, with converging total momenta. Then $G * m_\ell \to G * m$ in H^1. The fixed-space covariance operators $\mathcal{C}_{m_\ell}$ converge to $\mathcal{C}_m$ in Hilbert–Schmidt norm by (3.1). The tail s_j is the Hilbert–Schmidt distance of this operator to the set of operators of rank at most j. Indeed, the projection argument in Proposition 3.3 gives the lower bound for every such approximation, and truncating the spectral expansion after j terms attains it. As a distance to a fixed set, s_j is therefore Lipschitz in Hilbert–Schmidt norm. Likewise, d_j is Lipschitz with respect to the velocity's H^1 norm. Both sides of the estimates for atomic measures converge. Passing to the limit therefore preserves the estimates obtained from the positive atomic projections. □

4.3 Proof of Theorem 1.1

Proof Put $A = \sum_i \lambda_i^2 = 2E$ and $t_j = \sum_{i>j}\lambda_i^2 = s_j^2$. Positivity gives the product bound

$$(n+1)!F_{n+1} \ge A\prod_{j=1}^{n} t_j.$$

Indeed, the left-hand side is the sum of products of $n+1$ entries over ordered distinct indices. After removing any j indices, the sum of the remaining entries is at least t_j. We sum over the

last index first and apply this observation successively. Tonelli's theorem justifies the same argument for an infinite sequence.

Since $d_n \le d_j$ for $1 \le j \le n$, multiplication of (4.3)–(4.4) yields

$$d_n^{4n} \le 2 \cdot 8^{n-1}(n+1)!\, E^n \prod_{j=1}^{n} t_j \le 8^{n-1}\big((n+1)!\big)^2 E^{n-1} F_{n+1}.$$

If $E = 0$, the measure is zero and the assertion is immediate. □

Proposition 4.5 (Improved bound for measures supported on three points) *If m has at most three atoms, then*

$$d_2(m)^8 \le 2E(m)F_3(m).$$

Proof It suffices to consider three distinct atoms with positive weights. Write $m = p_1\delta_{x_1} + p_2\delta_{x_2} + p_3\delta_{x_3}$, $x_1 < x_2 < x_3$, $A = 1 - e^{-(x_2-x_1)}$, $B = 1 - e^{-(x_3-x_2)}$. Merging the smaller of the first two weights into the other gives a squared error of $\min(p_1, p_2)^2 A$. Merging the smaller of the last two gives $\min(p_2, p_3)^2 B$. Thus

$$d_2^4 \le p_1 p_2^2 p_3 AB = p_2\sqrt{F_3}, \qquad F_3 = (p_1 p_2 p_3 AB)^2.$$

Squaring and using $p_2^2 \le 2E$ proves the claim. □

4.4 Sharpness of the approximation exponent

Proposition 4.6 (Sharpness of the power $4n$) *For every $n \ge 1$, the exponent $4n$ in (1.2) cannot be replaced by $2p < 4n$ in an estimate with a uniform constant on a fixed nonzero energy window and the same functional F_{n+1}.*

Proof Take $m_h = \sum_{a=0}^{n} \delta_{ah}$, $0 < h \le 1$. The Gram determinant product gives

$$F_{n+1}(m_h) = \det(T_{m_h})^2 = (1 - e^{-h})^{2n}, \qquad E(m_h) \longrightarrow (n+1)^2/2.$$

For any positive measure ν on at most n points, one of the $n+1$ disjoint intervals $(ah - h/4, ah + h/4)$ contains no atom of ν. Let f be the triangular hat on that interval, equal to one at ah and zero at the endpoints. Its pairing and norm are

$$\langle G * (m_h - \nu), f\rangle_{H^1} = 1, \qquad \|f\|_{H^1}^2 = 8/h + h/6.$$

By continuity, the pairing follows from $\langle G(\cdot - x), f\rangle_{H^1} = f(x)$ for $f \in H^1$. An atom at an interval endpoint makes no contribution. Hence

$$d_n(m_h)^2 \ge \frac{h}{8 + h^2/6}.$$

Merging the last atom into the preceding one gives the upper bound $d_n(m_h)^2 \le 1 - e^{-h}$. Thus the squared distance is comparable to h, whereas F_{n+1} is comparable to h^{2n}. Consequently, d_n^{2p}/F_{n+1} diverges for $p < 2n$; the energy factor remains bounded above and below. □

5 Stability of multipeakons and estimates for exit times

We prove Theorem 1.2 by combining the spectral estimates with a scalar semimartingale comparison.

Lemma 5.1 (Exit estimates for a nonnegative semimartingale) *Let Y be a nonnegative continuous semimartingale with deterministic initial value, satisfying $\mathrm{d}Y = b_t\,\mathrm{d}t + \sigma_t \cdot \mathrm{d}W_t$ and*

$$|b_t| \le aY_t, \qquad |\sigma_t|^2 \le vY_t^2, \qquad a, v \ge 0.$$

Let τ be a stopping time such that $Y_\tau \ge b$ on $\{\tau < \infty\}$, where $b > 0$. Then

$$\mathbb{P}(\tau \le T) \le \min\{1, e^{aT} Y_0/b\}.$$

If $Y_0 > 0$ and $vT > 0$, then

$$\mathbb{P}(\tau \le T) \le \exp\left(-\frac{[\log(b/Y_0) - aT]_+^2}{2vT}\right).$$

If $0 < Y_0 < b$ and $a > 0$, then $\mathbb{E}\tau \ge a^{-1}\log(b/Y_0)$. If $Y_0 = 0$, then Y is identically zero almost surely.

Proof The process $e^{-at}Y_t$ is a nonnegative local supermartingale and hence a supermartingale. On $\{\tau \le T\}$ its maximum up to T is at least $e^{-aT}b$, so the maximal inequality gives the first bound. When $Y_0 = 0$, nonnegativity and the supermartingale property imply $Y_t = 0$ for every rational t, and then for all t by continuity.

To prove positivity when $Y_0 > 0$, set $\alpha_t = b_t/Y_t$ and $\beta_t = \sigma_t/Y_t$ on $\{Y_t > 0\}$, and set both ratios to zero on $\{Y_t = 0\}$. The coefficient bounds imply $|\alpha_t| \le a$, $|\beta_t|^2 \le v$ and $\mathrm{d}Y_t = \alpha_t Y_t\,\mathrm{d}t + Y_t\beta_t \cdot \mathrm{d}W_t$. The unique solution of this linear equation is

$$Y_t = Y_0 \exp\left\{\int_0^t \left(\alpha_s - \tfrac{1}{2}|\beta_s|^2\right)\mathrm{d}s + \int_0^t \beta_s \cdot \mathrm{d}W_s\right\}.$$

Thus Y stays positive on finite intervals, and

$$\log(Y_t/Y_0) \le at + N_t, \qquad N_t = \int_0^t \beta_s \cdot \mathrm{d}W_s, \qquad [N]_t \le vt.$$

If $h = \log(b/Y_0) - aT > 0$, the event $\{\tau \le T\}$ implies $\sup_{t\le T} N_t \ge h$. For every $\theta > 0$, the exponential martingale inequality gives $\mathbb{P}(\sup_{t\le T} N_t \ge h) \le \exp(-\theta h + \theta^2 vT/2)$. Choosing $\theta = h/(vT)$ proves the logarithmic bound; when $h \le 0$, it reduces to the bound by one.

Finally, assume $0 < Y_0 < b$ and put $\zeta = \inf\{t : Y_t \ge b\}$. Continuity gives $\zeta \le \tau$. If $\mathbb{E}\zeta = \infty$, the mean bound follows immediately. Otherwise $\zeta < \infty$ almost surely and $Y_\zeta = b$. Optional stopping of the nonnegative supermartingale at $\zeta \wedge t$, followed by Fatou's lemma, gives $b\mathbb{E}e^{-a\zeta} \le Y_0$. Jensen's inequality yields $\mathbb{E}\zeta \ge a^{-1}\log(b/Y_0)$, which proves the stated bound for $\mathbb{E}\tau$. □

5.1 Proof of Theorem 1.2

Proof The relative Green-operator estimates imply

$$|E_k'| \le 2\kappa_{1,k}E, \qquad |E_k''| \le 2(\kappa_{1,k}^2 + \kappa_{2,k})E.$$

Indeed, $E = \frac{1}{2}\operatorname{tr} T^2$, $E' = \operatorname{tr}(TT')$, and $E'' = \operatorname{tr}((T')^2) + \operatorname{tr}(TT'')$. Apply the relative factorizations in (3.3) and the Hilbert–Schmidt inequality $\|T^{1/2}KT^{1/2}\|_{\mathrm{HS}} \le \|K\|\,\|T\|_{\mathrm{HS}}$. Together with (3.5) for $r = n+1$, the product rule gives

$$|\mathcal{W}'_{n,k}| \le 4n\kappa_{1,k}\mathcal{W}_n,$$
$$|\mathcal{W}''_{n,k}| \le 4n((4n-1)\kappa_{1,k}^2 + \kappa_{2,k})\mathcal{W}_n.$$

The coefficient of κ_1^2 in the second derivative is

$$2(n-1)(2n-3) + 2(n+1)(2n+1) + 8(n-1)(n+1) = 4n(4n-1).$$

The exact energy and exterior-power balances yield

$$\mathrm{d}\mathcal{W}_n = \varepsilon \sum_k \mathcal{W}'_{n,k}\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k \mathcal{W}''_{n,k}\,\mathrm{d}t.$$

For each fixed n, these quantities are polynomials in the spectral traces. The total-momentum moment bounds therefore justify localization. Alternatively, nonnegativity turns the discounted local supermartingale into a supermartingale directly.

The spatial distance has continuous paths by the H^1 continuity of u. Apply Lemma 5.1 with $Y = \mathcal{W}_n$, $a = \gamma_n$, $v = \varepsilon^2 V_n^*$ and $b = b_{n,\eta}^*$. By the approximation theorem, an exit requires $Y \ge b$. The lemma therefore gives the three exit estimates. If $\gamma_n = 0$, the effective noise coefficients in the displayed derivative bounds vanish, and $\mathcal{W}_n$ is conserved. In particular, $\tau_{n,\eta}^* = \infty$ whenever $\mathcal{W}_n(m_0) < b_{n,\eta}^*$ in this case. If the initial value is zero, it remains zero by the supermartingale property, for arbitrary γ_n.

Finally, (3.6) with $r = n+1$ and $E(m_0) \le (B+\delta)^2$ imply

$$\mathcal{W}_n(m_0) \le \frac{2^{n+1}}{n!}(B+\delta)^{4n-2}\delta^2.$$

For $n = 2$, the refined bound is $\mathcal{W}_2(m_0) \le 2(B+\delta)^6\delta^2$. Substituting these bounds gives (1.4) and the improved coefficient for two peaks. □

Corollary 5.2 (A logarithmic extension of the noise time scale) *Fix $n \ge 1$, $\eta > 0$ and a nonnegative reference measure $\widetilde{m}_0$ with at most n atoms. Put $B = \|G * \widetilde{m}_0\|_{H^1}$ and*

$$\Gamma_n = 2n\sum_k \big((4n-1)\kappa_{1,k}^2 + \kappa_{2,k}\big).$$

Assume $\Gamma_n > 0$ and $\varepsilon > 0$, so that $\gamma_n = \varepsilon^2\Gamma_n$. For $0 < a < 2/\Gamma_n$ and $0 < \delta < 1$, set $T_\delta = a\varepsilon^{-2}\log(1/\delta)$ and define

$$K_{n,B,\eta} = \frac{c_n 2^{n+1}(B+1)^{4n-2}}{n!\,\eta^{4n}}.$$

For every finite nonnegative m_0 with $\|G * m_0 - G * \widetilde{m}_0\|_{H^1} \le \delta$,

$$\mathbb{P}(\tau_{n,\eta}^* \le T_\delta) \le \min\{1, K_{n,B,\eta}\delta^{\,2-a\Gamma_n}\} \longrightarrow 0 \qquad (\delta \downarrow 0).$$

The estimate is uniform over $\varepsilon > 0$ and finite nonnegative perturbations with no total-momentum bound.

Proof Since $e^{\gamma_n T_\delta} = \delta^{-a\Gamma_n}$ and $B + \delta \le B + 1$, substitution in (1.4) gives the stated bound. Its exponent is positive by the choice of a. □

For $n = 2$ the general coefficient is $c_2 = 288$, $\gamma_2 = \varepsilon^2 \sum_k (28\kappa_{1,k}^2 + 4\kappa_{2,k})$ and $V_2^* = 64 \sum_k \kappa_{1,k}^2$. If the initial measure has at most three atoms, this support class is invariant and Proposition 4.5 allows $c_2 = 2$ in all the exit estimates. In particular, the initial-error prefactor is 4, whereas it is 576 for arbitrary nonnegative finite momentum measures.

Remark 5.3 (Optimal quadratic initial-error power) For the functional F_{n+1}, the power δ^2 cannot be replaced by $\delta^{2+\alpha}$, for any $\alpha > 0$, with a uniform coefficient on a fixed nonzero energy window. Indeed, fix $x_1 < \cdots < x_n < q$ and $p_1, \ldots, p_n > 0$, and put

$$\widetilde{m} = \sum_{j=1}^{n} p_j \delta_{x_j}, \qquad m_a = \widetilde{m} + a\delta_q.$$

Then $\|G * m_a - G * \widetilde{m}\|_{H^1}^2 = a^2/2$, while the exponential-kernel Gram determinant, with $(z_1, \ldots, z_{n+1}) = (x_1, \ldots, x_n, q)$, gives

$$F_{n+1}(m_a) = a^2 \left(\prod_{j=1}^{n} p_j \right)^2 \prod_{j=1}^{n} (1 - e^{-(z_{j+1} - z_j)})^2.$$

Its coefficient of a^2 is strictly positive, and $E(m_a) \to E(\widetilde{m}) > 0$. The same conclusion holds for $\mathcal{W}_n = E^{n-1} F_{n+1}$. Hence the quadratic initial-error power in the functional estimate is sharp.

The coefficient 2 in $F_3 \le 2E^2\delta^2$ is sharp. For $p > 0$, take

$$\widetilde{m}_L = p\delta_0 + p\delta_L, \qquad m_{L,a} = \widetilde{m}_L + a\delta_{2L}.$$

Here $\delta^2 = a^2/2$ and $F_3(m_{L,a}) = p^4 a^2 (1 - e^{-L})^4$. Letting $L \to \infty$ and then $a \downarrow 0$ gives $F_3/(E^2\delta^2) \to 2$, also with E^2 replaced by $(\|G * \widetilde{m}_L\|_{H^1} + \delta)^4$.

6 Continuous dependence at multipeakon initial data

The proof of Theorem 1.3 has three steps. We first decompose the perturbed momentum into localized nonnegative components and a nonnegative remainder, with initial errors controlled by the H^1 distance. We then estimate their evolution under the full velocity and compare the component parameters with the specified reference driven by the same Brownian motions. Finally, we remove a stopping time based on local mass and remainder energy. The ordered-component calculations also give a comparison using total momentum, recorded separately in Appendix B.

Within this section, α_n is the growth coefficient for the comparison of specified solutions; the coefficient for approximation with prescribed total momentum continues to be denoted by a_n. The following quantitative version records the constants used in the proof.

Fix $n \ge 1$, deterministic $Q_1^0 < \cdots < Q_n^0$, and $P_i^0 > 0$. The specified reference is

$$v_t = \sum_{i=1}^{n} P_i(t) G(\cdot - Q_i(t)), \qquad P_t = \sum_i P_i(t),$$

where, on the same Brownian space,

$$\mathrm{d}Q_i = \sum_j P_j G(Q_i - Q_j)\,\mathrm{d}t + \varepsilon \sum_k \xi_k(Q_i) \circ \mathrm{d}W^k,$$
$$\mathrm{d}P_i = -P_i \sum_j P_j G'(Q_i - Q_j)\,\mathrm{d}t - \varepsilon P_i \sum_k \xi_k'(Q_i) \circ \mathrm{d}W^k.$$

The reference positions remain distinct and the weights positive by Proposition 2.3. Write $M_t = m_t(\mathbb{R})$ and $\delta = \|u_0 - v_0\|_{H^1}$. With the coefficient norms in (1.3), put

$$A_m = \frac{1}{2}\sum_k (L_{1,k}^2 + L_{0,k}L_{2,k}), \qquad V_m = \sum_k L_{1,k}^2.$$

The probability bound uses the following constants, determined by the noise norms:

$$b_k^{\mathcal{R}} = 2(L_{0,k} + L_{1,k}), \qquad c_k^{\mathcal{R}} = (b_k^{\mathcal{R}})^2 + 2L_{0,k}L_{1,k} + 2L_{0,k}L_{2,k}.$$

Set

$$b_E = \frac{\varepsilon_*^2}{2}\sum_k c_k^{\mathcal{R}}, \qquad v_E = \varepsilon_*^2 \sum_k (b_k^{\mathcal{R}})^2. \tag{6.1}$$

Theorem 6.1 (Quantitative continuous dependence at multipeakon initial data) *Fix the specified positive multipeakon above and $T > 0$. For $n \geq 2$, set $r = \min\{\frac{1}{2}\min_i(Q_{i+1}^0 - Q_i^0), 1\}$; for $n = 1$, put $r = 1$. Write*

$$H^2 = 2/r + 2r/3, \quad \Lambda = 1 + 1/r, \quad \delta_{\mathrm{en}} = (2\Lambda)^{-1}, \quad S_* = 2P_0 + \sqrt{n}H.$$

Let m_0 be any deterministic finite nonnegative measure with $\delta = \|G * m_0 - v_0\|_{H^1} \leq \delta_{\mathrm{en}}$. There is no upper bound on $m_0(\mathbb{R})$ and no total-momentum convergence assumption. Put $\mathcal{X}_T = \sup_{t\leq T}\|u_t - v_t\|_{H^1}$. For every $R > \max\{1, S_*\}$ and $\eta > 0$,

$$\mathbb{P}(\mathcal{X}_T \geq \eta) \leq \min\left\{1,\ \mathcal{B}_R(T) + \mathcal{D}_R(T)\Lambda^2\delta^2 + \frac{C_T R^2 e^{\alpha_n R T}\delta}{\eta^2}\right\}. \tag{6.2}$$

Here $\alpha_n = 4(2\sqrt{n}+n+\frac{1}{2})+2$, C_T is independent of $R, m_0(\mathbb{R})$ and δ, and the explicit constants are given in (6.25). *With b_E, v_E from* (6.1),

$$\mathcal{D}_R(T) = 2\exp\{(2R + b_E + v_E/2)T\},$$
$$\mathcal{B}_R(T) = \begin{cases} \exp\left[-\dfrac{[\log(R/S_*) - (2^{-1/2} + \varepsilon^2 A_m)T]_+^2}{2\varepsilon^2 V_m T}\right], & \varepsilon^2 V_m > 0, \\ 0, & \varepsilon^2 V_m = 0,\ R > S_* e^{T/\sqrt{2}}, \\ 1, & \varepsilon^2 V_m = 0,\ R \leq S_* e^{T/\sqrt{2}}. \end{cases} \tag{6.3}$$

For every sequence of deterministic nonnegative finite initial momenta with $\delta \to 0$, the corresponding solutions converge to v in $L^p(\Omega; C([0,T]; H^1))$ for each $1 \leq p < \infty$. If $\varepsilon^2 V_m = 0$, one has the stronger bound $\mathbb{E}\mathcal{X}_T^2 \leq C\delta$ without stopping. If $\varepsilon^2 V_m > 0$, then for every $0 < c < (2\varepsilon^2 V_m T)^{-1}$ and sufficiently small $\delta > 0$,

$$\mathbb{E}\mathcal{X}_T^2 \leq C_c \exp\{-c[\log\log(1/\delta)]^2\}. \tag{6.4}$$

The constants may depend on the fixed reference, T, ε_* and the noise norms, but not on the total momentum, number of atoms or spatial moments of the perturbation. The estimates use neither a stopping time based on the evolving gaps nor a lower bound for the reference weights. In (6.4)*, ε is fixed and C_c may additionally depend on c and ε.*

6.1 Local decomposition and initial estimates

The full momentum of a nonnegative measure need not be controlled by the H^1 norm of its velocity. We isolate the momentum near the reference peaks and measure the remaining part by its velocity energy. Throughout the proof, $n_0 = \sum_{i=1}^n P_i^0 \delta_{Q_i^0}$ and $\delta = \|u_0 - v_0\|_{H^1}$.

The characteristic representation of Theorem 2.2 has the form

$$m_t = X_{t\#}(p_t m_0), \qquad \mathrm{d}X_t(x) = u_t(X_t(x))\,\mathrm{d}t + \varepsilon \sum_k \xi_k(X_t(x)) \circ \mathrm{d}W_t^k, \tag{6.5}$$

where X_t is strictly increasing, $X_0(x) = x$, and, for m_0-almost every label,

$$\mathrm{d}p_t(x) = -p_t(x)\bar{u}_x(t, X_t(x))\,\mathrm{d}t - \varepsilon p_t(x) \sum_k \xi_k'(X_t(x)) \circ \mathrm{d}W_t^k, \qquad p_0 = 1.$$

We use the canonical position flow defined for all labels. Since the full velocity is spatially Lipschitz, this flow is defined at each reference point Q_i^0 even when m_0 has no atom there. Set $K(x,q) = 1 - e^{-|x-q|}$.

For $n \geq 2$ put $r = \min\{\frac{1}{2}\min_i(Q_{i+1}^0 - Q_i^0), 1\}$; for $n = 1$ put $r = 1$. Define

$$h_i(x) = \left(1 - \frac{|x - Q_i^0|}{r}\right)_+, \qquad \chi(x) = 1 - \sum_{i=1}^n h_i(x).$$

The supports of the h_i have disjoint interiors. In particular, $0 \leq \chi \leq 1$, $\chi \in W^{1,\infty}(\mathbb{R})$ and $\|\chi'\|_\infty \leq r^{-1}$. Set

$$m_i^{c,0} = h_i m_0, \qquad m^{c,0} = \sum_i m_i^{c,0}, \qquad \nu_0 = \chi m_0. \tag{6.6}$$

We transport all these measures using the same characteristic representation of Theorem 2.2. Denoting the position flow and weight by X_t, p_t, we obtain

$$m_i^c(t) = X_{t\#}(p_t m_i^{c,0}), \qquad m_t^c = \sum_i m_i^c(t), \qquad \nu_t = X_{t\#}(p_t \nu_0),$$

$$c_t = G * m_t^c, \qquad w_t = G * \nu_t, \qquad u_t = c_t + w_t, \qquad \mathcal{R}_t = \|w_t\|_{H^1}^2,$$

$$M_i^c(t) = m_i^c(t)(\mathbb{R}), \qquad M_t^c = \sum_i M_i^c(t), \qquad q_i(t) = X_t(Q_i^0),$$

$$L_i^c(t) = \int_{\mathbb{R}} K(x, q_i(t)) m_i^c(t, \mathrm{d}x), \qquad L_t = \sum_i L_i^c(t).$$

Applying the characteristic product formula to the fixed initial multipliers h_i and χ shows that each m_i^c and ν satisfies the linear weak momentum equation with the full velocity u and its averaged derivative. The component equations therefore retain the interactions with all other components. No weight value is needed at a reference point not charged by m_0. Distinct localized measures do not share an atomic label and retain their strict spatial order. An initial atom may be divided between one localized measure and the remainder; both pieces then use the same characteristic position and weight. Only the localized measures enter the order estimates below.

Lemma 6.2 (Initial estimates from the velocity error) *Put*

$$H^2 = \frac{2}{r} + \frac{2r}{3}, \qquad J^2 = 2r + \frac{2r^3}{3}, \qquad \Lambda = 1 + \frac{1}{r}.$$

Then

$$\mathcal{R}_0 \le \Lambda^2 \delta^2, \tag{6.7}$$

$$\sum_i |M_i^c(0) - P_i^0|^2 \le H^2 \delta^2, \qquad L_0 \le \sqrt{n}\, J\delta,$$

$$S_0^c := M_0^c + P_0 \le 2P_0 + \sqrt{n}\, H\delta. \tag{6.8}$$

Combining the three squared initial errors gives

$$\mathcal{R}_0 + L_0^2 + \sum_i |M_i^c(0) - P_i^0|^2 \le (\Lambda^2 + nJ^2 + H^2)\delta^2. \tag{6.9}$$

None of these estimates depends on $m_0(\mathbb{R})$.

Proof For a finite signed measure λ, the dual norm associated with our H^1 inner product obeys $\|\lambda\|_{H^{-1}} = \|G * \lambda\|_{H^1}$. Multiplication by a bounded Lipschitz function χ has operator norm at most $\|\chi\|_\infty + \|\chi'\|_\infty$ on H^1, since

$$\|\chi f\|_{H^1} \le \|\chi\|_\infty \|f\|_{H^1} + \|\chi'\|_\infty \|f\|_{L^2}.$$

By duality, the same multiplier bound holds on H^{-1}. Because $\chi(Q_i^0) = 0$,

$$\nu_0 = \chi(m_0 - n_0), \qquad \|w_0\|_{H^1} \le \Lambda \|u_0 - v_0\|_{H^1}.$$

This gives (6.7) without a mass estimate for the remainder.

The h_i/H are orthonormal in H^1 and $h_i(Q_j^0) = 1_{i=j}$. Thus

$$M_i^c(0) - P_i^0 = \langle u_0 - v_0, h_i \rangle_{H^1},$$

and Bessel's inequality proves the weight estimate. Summing the weight differences and applying Cauchy–Schwarz gives (6.8).

Let $\phi_i(x) = K(x, Q_i^0) h_i(x)$. This function vanishes at every reference node, so $L_i^c(0) = \langle u_0 - v_0, \phi_i \rangle_{H^1}$. The ϕ_i have disjoint supports up to endpoints. On each half of its support, ϕ_i' is the difference, up to an overall sign, of two nonnegative terms bounded by one. Hence $|\phi_i'| \le 1$ almost everywhere, while $|\phi_i(x)| \le |x - Q_i^0|$. Therefore

$$\|\phi_i\|_{H^1}^2 \le 2r + 2r^3/3 = J^2.$$

Bessel's inequality followed by Cauchy–Schwarz gives $L_0 \le \sqrt{n}\, J\delta$. □

6.2 Ordered-component identities and auxiliary full-mass estimates

The characteristic product formula applies to components obtained by fixed nonnegative multipliers of m_0. We use an exhaustive partition for the auxiliary estimates in Appendix B; Lemma 6.8 gives the localized residual estimates.

Partition the line into the Voronoi intervals of Q_i^0, assigning each midpoint to the interval on its left. For $n = 1$ use $I_1 = \mathbb{R}$. An initial atom is assigned wholly to one interval. Let $m_i^0 = 1_{I_i} m_0$.

$$m_i(t) = X_{t\#}(p_t m_i^0), \qquad q_i(t) = X_t(Q_i^0),$$
$$M_i(t) = m_i(t)(\mathbb{R}), \qquad L_i(t) = \int_{\mathbb{R}} K(x, q_i(t)) m_i(t, \mathrm{d}x).$$

The component masses are nonnegative, and zero masses are allowed. The points q_i are strictly ordered even when $m_i = 0$. Write $L = \sum_i L_i$. The component measures sum to m, and $\sum_i M_i = M$.

In this subsection, m_i, M_i, L_i and $L = \sum_i L_i$ refer to the exhaustive partition. The main comparison uses the localized quantities m_i^c, M_i^c, L_i^c and reuses L for their summed concentration. Both constructions use the same transported reference points q_i.

For the auxiliary estimates only, put

$$\theta_R = \inf\{t \geq 0 : M_t + P_t \geq R\}, \qquad R > M_0 + P_0.$$

In every stopped supremum, both solutions are evaluated at the same stopped time. This full-mass stopping time is distinct from σ_R in (6.18); no bound on M_0 is imposed in the main theorem.

Lemma 6.3 (Group equation and order) *Each m_i is adapted and narrowly continuous. For every $f \in C_c^\infty(\mathbb{R}) \cup \{1\}$, it satisfies*

$$\mathrm{d}\langle f, m_i\rangle = \langle uf' - \bar{u}_x f, m_i\rangle \,\mathrm{d}t + \varepsilon \sum_k \langle B_k f, m_i\rangle \,\mathrm{d}W^k$$
$$+ \frac{\varepsilon^2}{2} \sum_k \langle B_k^2 f, m_i\rangle \,\mathrm{d}t, \qquad u = G * m, \quad \bar{u}_x = G' * m.$$

This equation is linear in the component measure when the full velocity u, generated by m, is fixed. For $i < j$, pairs drawn from $m_i \otimes m_j$ satisfy $x < y$ almost everywhere, while $q_i < q_j$.

Proof Apply the characteristic product formula to $p_t(x) f(X_t(x))$ and integrate over the fixed Borel set I_i. The essential upper and lower weight bounds in Theorem 2.2 justify stochastic Fubini after localization. The total mass bounds then remove this auxiliary localization. Since the measures are finite, spatial cutoffs require no spatial moment assumption. Dominated convergence on the localized characteristic representation gives narrow continuity. For different intervals, all initial labels are strictly ordered; strict monotonicity of X_t preserves this order. The midpoint convention ensures that no common atomic label is divided between two groups. Summing their weak equations recovers that of m. □

Lemma 6.4 (Concentration balance) *Each L_i is a continuous nonnegative semimartingale with*

$$\mathrm{d}L_i = R_i\,\mathrm{d}t + \varepsilon \sum_k a_{ik}\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k c_{ik}\,\mathrm{d}t,$$

where

$$\begin{aligned} R_i &= \int_{\mathbb{R}} \Big[\operatorname{sgn}(x-q_i)e^{-|x-q_i|}(u(x)-u(q_i)) - \bar{u}_x(x)K(x,q_i)\Big]\,\mathrm{d}m_i(x),\\ a_{ik} &= \int_{\mathbb{R}} \mathcal{S}_k K(x,q_i)\,\mathrm{d}m_i(x), \qquad c_{ik} = \int_{\mathbb{R}} \mathcal{S}_k^2 K(x,q_i)\,\mathrm{d}m_i(x),\\ \mathcal{S}_k &= \xi_k(x)\partial_x + \xi_k(q_i)\partial_{q_i} - \xi_k'(x). \end{aligned}$$

These coefficients satisfy

$$|R_i| \le ML_i, \quad |a_{ik}| \le 2L_{1,k}L_i, \quad |c_{ik}| \le (6L_{1,k}^2 + 2L_{0,k}L_{2,k})L_i. \tag{6.10}$$

Consequently

$$\mathbb{E}\sup_{t\le T} L_{t\wedge\theta_R}^2 \le C_L L_0^2, \quad C_L = 2\exp\left\{\left[2R + \varepsilon_*^2\sum_k (18L_{1,k}^2 + 2L_{0,k}L_{2,k})\right]T\right\}. \tag{6.11}$$

Proof For a component transported by the full velocity, set $s = \operatorname{sgn}(x-q)$, $e = e^{-|x-q|}$, $a = \xi(x) - \xi(q)$, $c = \xi'(x)$, $b = \xi(x)\xi'(x) - \xi(q)\xi'(q)$ and $h = \xi(x)\xi''(x)$,

$$\mathcal{S}_\xi K = sae - cK, \qquad \mathcal{S}_\xi^2 K = e\{s(b-2ca) - a^2\} + (c^2 - h)K.$$

Both expressions extend continuously by zero to $x = q$. The inequalities $re^{-r} \le 1 - e^{-r}$ and $r^2e^{-r} \le 2(1-e^{-r})$ give the noise bounds. The deterministic estimate uses

$$|u(x) - u(q)| \le \tfrac{M}{2}K(x,q), \qquad |\bar{u}_x| \le M/2.$$

To justify the weak formula, use an even regularization of K and apply the component equation jointly with the SDE for q_i. The singular second derivative is multiplied by $(\xi_k(x) - \xi_k(q))^2$, so its regularized contribution is uniformly $O(\rho)$ at mollifier scale ρ. The deterministic first derivative is multiplied by $u(x) - u(q)$. Thus no local-time term remains. The stochastic Fubini theorem and the mass moment bounds justify passage to the limit. Equivalently, distinct characteristic labels never meet, and a label equal to Q_i^0 contributes zero identically. This also treats a possible atom at q_i.

Summing (6.10) gives the same relative bounds for L. Write its stopped equation as $\mathrm{d}L = r_t L\,\mathrm{d}t + L\gamma_t \cdot \mathrm{d}W$, setting the ratios to zero on $\{L = 0\}$. Then

$$r_t \le R + \varepsilon_*^2\sum_k (3L_{1,k}^2 + L_{0,k}L_{2,k}), \qquad |\gamma_t|^2 \le 4\varepsilon_*^2\sum_k L_{1,k}^2.$$

The coefficients vanish when $L = 0$, so the stochastic exponential representation is valid also when $L_0 = 0$. For $|\gamma|^2 \le \Gamma^2$, $\mathbb{E}\sup_{t\le T}\mathcal{E}(2\gamma\cdot W)_t \le 2e^{2\Gamma^2 T}$ by Cauchy–Schwarz and the L^2 Doob inequality applied to $\mathcal{E}(2\gamma\cdot W)$, whose second moment at the terminal time is at most $e^{4\Gamma^2 T}$. Applying this to L^2 gives (6.11). □

Equations for component weights and positions.

Define $h_k = (\xi_k')^2 - \xi_k\xi_k''$ and $f_k = \xi_k\xi_k'$. For an ordered position vector q and nonnegative weight vector a, put

$$U_i(q,a) = \sum_j a_j G(q_i - q_j), \qquad V_i(q,a) = \sum_j a_j G'(q_i - q_j).$$

Lemma 6.5 (Residual equations) *The component masses M_i and points q_i obey*

$$\begin{aligned} \mathrm{d}q_i &= \left[U_i(q, M.) + r_i^q + \frac{\varepsilon^2}{2} \sum_k f_k(q_i) \right] \mathrm{d}t + \varepsilon \sum_k \xi_k(q_i)\, \mathrm{d}W^k, \\ \mathrm{d}M_i &= \left[-M_i V_i(q, M.) + r_i^M + \frac{\varepsilon^2}{2} \sum_k (M_i h_k(q_i) + e_{ik}^2) \right] \mathrm{d}t - \varepsilon \sum_k (M_i \xi_k'(q_i) + e_{ik}^1)\, \mathrm{d}W^k. \end{aligned} \tag{6.12}$$

Let

$$\begin{aligned} H_{0,k} &= L_{1,k}^2 + L_{0,k}L_{2,k}, & H_{1,k} &= L_{1,k}L_{2,k} + L_{0,k}L_{3,k}, \\ c_{1,k} &= 2L_{2,k} + 4L_{1,k}, & c_{2,k} &= 2H_{1,k} + 4H_{0,k}. \end{aligned}$$

Then

$$|r_i^q| \le L/2, \quad \sum_i |r_i^M| \le ML, \quad |e_{ik}^1| \le c_{1,k}L_i, \quad |e_{ik}^2| \le c_{2,k}L_i. \tag{6.13}$$

Proof The inequality $|e^{-|a|} - e^{-|b|}| \le 1 - e^{-|a-b|}$ implies

$$|G(s-x) - G(s-y)| \le K(x,y)/2.$$

Integrating against the groups gives the bound for r_i^q. For $i \ne j$, group order gives the same sign for $x - y$ and $q_i - q_j$. Hence, almost everywhere with respect to $m_i \otimes m_j$,

$$|G'(x-y) - G'(q_i - q_j)| \le \tfrac{1}{2}\{K(x,q_i) + K(y,q_j)\}. \tag{6.14}$$

Within one group the double integral of $G'(x-y)$ is zero by antisymmetry, including the diagonal. Applying the test 1 to the group equation, canceling this self-interaction, and then using (6.14) gives

$$|r_i^M| \le \tfrac{1}{2} \sum_{j \ne i} (M_j L_i + M_i L_j).$$

Summing gives a bound of ML. We use this cancellation before comparing the components, since G' has no pointwise Lipschitz bound within one component.

For a bounded Lipschitz function f and a positive finite measure ν,

$$\left| \int_{\mathbb{R}} (f(x) - f(q))\, \mathrm{d}\nu(x) \right| \le (2\,\mathrm{Lip}(f) + 4\|f\|_\infty) \int_{\mathbb{R}} K(x,q)\, \mathrm{d}\nu(x).$$

To prove this inequality, we split the integral into the regions $|x - q| \le 1$ and its complement. Use $f = \xi_k'$ and $f = h_k$, noting $h_k' = \xi_k'\xi_k'' - \xi_k\xi_k'''$. The errors in (6.12) are exactly these averaging errors. This proves the residual bounds and the full Itô equations, including the momentum correction $M_i h_k/2$. □

Profile reconstruction.

Lemma 6.6 (Approximation by point masses at transported reference points) *With the notation above,*

$$\|u - \sum_i M_i G(\cdot - q_i)\|_{H^1}^2 \le ML. \tag{6.15}$$

Before θ_R one also has

$$\|u - v\|_{H^1}^2 \le 2ML + 2n|M_\cdot - P_\cdot|_2^2 + 4R^2|q - Q|_2. \tag{6.16}$$

Proof For a positive measure ν of mass a,

$$\|G * \nu - aG(\cdot - z)\|_{H^1}^2 = a\int_{\mathbb{R}} K(x,z)\,\mathrm{d}\nu(x) - \left(\tfrac{a^2}{2} - \|G*\nu\|_{H^1}^2\right) \le a\int_{\mathbb{R}} K(x,z)\,\mathrm{d}\nu(x).$$

The quantity in parentheses is nonnegative. Summing the resulting norms and using $(\sum_i \sqrt{M_i L_i})^2 \le ML$ gives (6.15). The reproducing identity for G gives

$$\|G(\cdot - q) - G(\cdot - Q)\|_{H^1}^2 = 1 - e^{-|q-Q|} \le |q - Q|.$$

Separate the weight changes from the position changes. Their norms are at most $\sqrt{n/2}|M_\cdot - P_\cdot|_2$ and $R\sqrt{\max_i |q_i - Q_i|}$, respectively. The squared norm of their sum is bounded by twice the sum of the squared norms. The final triangle inequality gives (6.16). □

6.3 Energy and forcing from the remainder

We return to the localized split $m_t = m_t^c + \nu_t$ in (6.6). In the remainder of the proof, $L = \sum_i L_i^c$ denotes the local concentration error. The estimates below use M^c, the reference momentum and the energy $\mathcal{R}$; the full momentum of the perturbation does not enter the comparison threshold.

Lemma 6.7 (Energy of the transported remainder) *Let $\bar{w}_x = G' * \nu$ and $\bar{c}_x = G' * m^c$, retaining $G'(0) = 0$. The remainder energy has the exact Itô equation*

$$\mathrm{d}\mathcal{R} = R_{\mathcal{R}}\,\mathrm{d}t + \varepsilon\sum_k A_k^{\mathcal{R}}\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k C_k^{\mathcal{R}}\,\mathrm{d}t,$$

where

$$R_{\mathcal{R}} = 2\int_{\mathbb{R}} (u\bar{w}_x - \bar{u}_x w)\,\mathrm{d}\nu = 2\int_{\mathbb{R}} (c\bar{w}_x - \bar{c}_x w)\,\mathrm{d}\nu, \qquad |R_{\mathcal{R}}| \le 2M^c\mathcal{R}.$$

For

$$\mathcal{T}_\xi = \xi(x)\partial_x + \xi(y)\partial_y - \xi'(x) - \xi'(y),$$

one has

$$A_k^{\mathcal{R}} = \iint_{\mathbb{R}^2} \mathcal{T}_{\xi_k} G(x-y)\,\mathrm{d}\nu(x)\,\mathrm{d}\nu(y), \qquad C_k^{\mathcal{R}} = \iint_{\mathbb{R}^2} \mathcal{T}_{\xi_k}^2 G(x-y)\,\mathrm{d}\nu(x)\,\mathrm{d}\nu(y).$$

The kernels are defined on the diagonal by continuous extension. With $b_k^{\mathcal{R}}$ and $c_k^{\mathcal{R}}$ defined before Theorem 6.1, one has

$$|A_k^{\mathcal{R}}| \le b_k^{\mathcal{R}}\mathcal{R}, \qquad |C_k^{\mathcal{R}}| \le c_k^{\mathcal{R}}\mathcal{R}. \tag{6.17}$$

Proof Applying the tensor Itô formula to the linear equation for ν with kernel $G(x-y)$ gives the first expression for $R_{\mathcal{R}}$. Substituting $u = c + w$ cancels the term $w\bar{w}_x - \bar{w}_x w$ pointwise, also at atoms under the averaged-derivative convention. By positivity, $|\bar{w}_x| \le w$ and $|\bar{c}_x| \le c \le M^c/2$. These inequalities give the drift bound.

For one field put $s = \operatorname{sgn}(x-y)$, $a = \xi(x)-\xi(y)$ and $b = \xi'(x)+\xi'(y)$. Direct differentiation away from the diagonal gives

$$\begin{aligned}\mathcal{T}_\xi G &= (-sa-b)G,\\ \mathcal{T}_\xi^2 G &= \big[(sa+b)^2 - s\{\xi(x)\xi'(x)-\xi(y)\xi'(y)\} - \xi(x)\xi''(x)-\xi(y)\xi''(y)\big]G.\end{aligned}$$

The second bracket has diagonal limit $4(\xi')^2 - 2\xi\xi''$. Using $|a| \le 2L_0$, $|b| \le 2L_1$ proves (6.17).

It remains to justify the measure decomposition and the energy identity. The initial multipliers in (6.6) are fixed bounded Borel functions. Applying the characteristic product formula in Theorem 2.2 to restricted or weighted initial labels gives their linear weak equations. This argument requires no second derivatives of the Lipschitz cutoffs. We first prove the tensor formula for finite sums of smooth product tests. Even regularization of G produces the only singular correction $[\xi(x)-\xi(y)]^2\rho_\eta(x-y)$, whose supremum is $O(\eta)$. All other terms converge to the displayed kernels, including their atomic diagonal values. The deterministic drift converges by bounded convergence and the averaged-derivative convention. For each individual solution, the mass and weight localization used for weak balances in Part I [22] justifies stochastic Fubini and passage to the limit in the regularization. This auxiliary full-mass localization is then removed; the quantitative bounds depend on M^c and $\mathcal{R}$, independently of the full momentum. No uniform bound on the initial full mass is required. □

For the remaining bounds, assume $\delta \le \min\{1, (2\Lambda)^{-1}\}$, choose $R > S_0^c$, and set

$$S_t^c = M_t^c + P_t, \qquad \sigma_R = \inf\{t \ge 0 : S_t^c \ge R \text{ or } \mathcal{R}_t \ge 1\}. \tag{6.18}$$

At every stopped time, continuity gives $S^c \le R$ and $\mathcal{R} \le 1$. In particular,

$$\|\bar{u}_x\|_\infty \le \|u\|_\infty \le \frac{M^c}{2} + \frac{\sqrt{\mathcal{R}}}{\sqrt{2}} \le B_R := \frac{R}{2} + \frac{1}{\sqrt{2}}.$$

The full measure remains finite; its total mass does not enter this stopping time.

Lemma 6.8 (Concentration and the contribution of the remainder) *The concentration L_i^c has the same noise coefficients as in Lemma 6.4, with m_i^c in place of m_i and the full velocity u. Its deterministic drift satisfies*

$$|R_i^c| \le 2\|\bar{u}_x\|_\infty L_i^c. \tag{6.19}$$

Consequently

$$\mathbb{E}\sup_{t\le T} L^2_{t\wedge\sigma_R} \le C_L^c L_0^2,$$

$$C_L^c = 2\exp\left\{\left[4B_R + \varepsilon_*^2 \sum_k (18L_{1,k}^2 + 2L_{0,k}L_{2,k})\right]T\right\}, \tag{6.20}$$

$$\mathbb{E}\sup_{t\le T} \mathcal{R}_{t\wedge\sigma_R} \le C_{\mathcal{R}}\mathcal{R}_0, \qquad C_{\mathcal{R}} = 2\exp\left\{\left[2R + \frac{\varepsilon_*^2}{2}\sum_k \{c_k^{\mathcal{R}} + (b_k^{\mathcal{R}})^2\}\right]T\right\}. \tag{6.21}$$

Write $f_k = \xi_k \xi_k'$ and $h_k = (\xi_k')^2 - \xi_k \xi_k''$. The exact equations for the localized weights and the points q_i are

$$\begin{aligned}
\mathrm{d}q_i &= \left[U_i(q, M^c_\cdot) + r_i^q + w(q_i) + \frac{\varepsilon^2}{2} \sum_k f_k(q_i) \right] \mathrm{d}t + \varepsilon \sum_k \xi_k(q_i)\, \mathrm{d}W^k, \\
\mathrm{d}M_i^c &= \left[-M_i^c V_i(q, M^c_\cdot) + r_i^M - \int_{\mathbb{R}} \bar{w}_x \, \mathrm{d}m_i^c \right] \mathrm{d}t \\
&\quad + \frac{\varepsilon^2}{2} \sum_k (M_i^c h_k(q_i) + e_{ik}^2)\, \mathrm{d}t - \varepsilon \sum_k (M_i^c \xi_k'(q_i) + e_{ik}^1)\, \mathrm{d}W^k.
\end{aligned} \tag{6.22}$$

The constants $c_{1,k}, c_{2,k}$ of Lemma 6.5 give

$$|r_i^q| \le L/2, \quad \sum_i |r_i^M| \le M^c L, \quad |e_{ik}^1| \le c_{1,k} L_i^c, \quad |e_{ik}^2| \le c_{2,k} L_i^c. \tag{6.23}$$

The additional drift terms contributed by the remainder satisfy

$$|w(q_i)| \le \frac{\sqrt{\mathcal{R}}}{\sqrt{2}}, \qquad \left| \int_{\mathbb{R}} \bar{w}_x \, \mathrm{d}m_i^c \right| \le \frac{M_i^c \sqrt{\mathcal{R}}}{\sqrt{2}}. \tag{6.24}$$

Proof In the concentration drift, put $d = |x - q_i|$. Since the full positive velocity is Lipschitz,

$$e^{-d} |u(x) - u(q_i)| \le \|\bar{u}_x\|_\infty d e^{-d} \le \|\bar{u}_x\|_\infty K(x, q_i).$$

The reaction term contributes the same bound, proving (6.19). The noise bounds for the concentration are unchanged, so the stochastic-exponential estimate in Lemma 6.4 proves (6.20).

Before the common stopping time, the drift ratio of the remainder energy equation is at most $2R + \varepsilon_*^2 \sum_k c_k^{\mathcal{R}}/2$, and its squared diffusion ratio is at most $\varepsilon_*^2 \sum_k (b_k^{\mathcal{R}})^2$. If $\mathcal{R}_0 > 0$, write the scalar solution as its stochastic exponential times its drift exponential. For a bounded integrand γ, Cauchy–Schwarz and Doob's L^2 maximal inequality give

$$\mathbb{E} \sup_{t \le T} \mathcal{E}(\gamma \cdot W)_t \le 2 \exp\left(\frac{T}{2} \sup |\gamma|^2 \right).$$

This proves (6.21). If $\mathcal{R}_0 = 0$, the relative coefficient bounds imply that the scalar solution vanishes identically.

Apply the linear equation for m_i^c to the test 1. Cancel interactions within each localized measure and compare interactions between distinct localized measures using their fixed order, exactly as in Lemma 6.5. The remaining velocity is w, and this gives (6.22) and (6.23). Finally, $|\bar{w}_x| \le w$ and the Green evaluation bound $\|w\|_\infty \le \|w\|_{H^1}/\sqrt{2}$ give (6.24). No derivative-kernel comparison with the remainder is required. □

The initial estimates in (6.9) and the two maximal estimates control the forcing in the parameter comparison. The comparison uses the common stopping time (6.18), whose threshold involves local mass and remainder energy. We now normalize the weight coordinates to keep the comparison exponent linear in that threshold.

6.4 Normalization of the group comparison

We compare positions and weights in the scaled state $z = (q-Q, (M_\cdot^c - P_\cdot)/R)$. With this scaling, the choice $R_\delta \asymp \log(1/\delta)$ balances removal of the stopping time against decay of the stopped error. We first record the martingale estimate used in the comparison. We take $b_{\mathrm{B}} = 8$ as an admissible universal constant in the continuous-martingale BDG inequality $\mathbb{E}\sup_{s\le t}|N_s| \le b_{\mathrm{B}}\mathbb{E}[N]_t^{1/2}$ for $N_0 = 0$. Write $N^* = \sup_{s\le t}|N_s|$. A stopped L^2 maximal inequality applied after the first crossing of λ, with the subsequent quadratic variation stopped at $\lambda^2/4$, gives

$$\mathbb{P}(N^* > 2\lambda) \le \mathbb{P}([N]_t^{1/2} > \lambda/2) + \tfrac{1}{4}\mathbb{P}(N^* > \lambda).$$

Integration over $\lambda > 0$ proves the choice $b_{\mathrm{B}} = 8$. Bounded stopping times and Fatou's lemma justify this argument.

Lemma 6.9 (A normalized forced comparison) *Let z be an adapted continuous vector semimartingale with deterministic initial value z_0. Suppose that, up to a stopping time τ, it obeys $\mathrm{d}z = (b+f)\,\mathrm{d}t + (\sigma+g)\,\mathrm{d}W$, where*

$$|b| \le \ell_b|z|, \quad \|\sigma\|_{\mathrm{HS}} \le \ell_\sigma|z|, \quad |f| \le BF, \quad \|g\|_{\mathrm{HS}} \le SF.$$

The coefficients are progressively measurable, the constants in these bounds are deterministic and nonnegative, and F is nonnegative and progressively measurable. We may first localize the integrals below to ensure integrability. For $Z(t) = \mathbb{E}\sup_{s\le t}|z_{s\wedge\tau}|^2$, one has

$$Z(t) \le 2|z_0|^2 + k_1\int_0^t Z(s)\,\mathrm{d}s + k_2\int_0^t \mathbb{E}F_{s\wedge\tau}^2\,\mathrm{d}s,$$
$$k_1 = 4\ell_b + 2 + (4+8b_{\mathrm{B}}^2)\ell_\sigma^2, \qquad k_2 = 2B^2 + (4+8b_{\mathrm{B}}^2)S^2.$$

Proof Itô's formula gives a drift bounded by $(2\ell_b + 1 + 2\ell_\sigma^2)|z|^2 + (B^2+2S^2)F^2$. The stopped martingale is

$$N_t = 2\int_0^{t\wedge\tau} \langle z_s, (\sigma_s+g_s)\,\mathrm{d}W_s\rangle.$$

The BDG and Young inequalities give

$$\mathbb{E}\sup_{s\le t}|N_s| \le \tfrac{1}{2}Z(t) + 2b_{\mathrm{B}}^2\mathbb{E}\int_0^{t\wedge\tau}\|\sigma+g\|_{\mathrm{HS}}^2\,\mathrm{d}s \le \tfrac{1}{2}Z(t) + 4b_{\mathrm{B}}^2\int_0^t(\ell_\sigma^2 Z(s) + S^2\mathbb{E}F_{s\wedge\tau}^2)\,\mathrm{d}s.$$

Absorbing $Z(t)/2$ gives the asserted inequality. Additional bounded stopping times justify the martingale estimates; Fatou's lemma then removes this localization. This proves the lemma. □

For the localized measures, we use

$$z = (q-Q, (M_\cdot^c - P_\cdot)/R), \qquad Y^{\mathrm{sc}} = |z|^2, \qquad F = L + \sqrt{\mathcal{R}}.$$

Recall $H_{0,k}, H_{1,k}, c_{1,k}, c_{2,k}$ from Lemma 6.5. Set

$$\kappa_n = 2\sqrt{n} + n + \tfrac{1}{2}, \qquad b_0 = \tfrac{\varepsilon_*^2}{2}\sum_k (2H_{0,k} + H_{1,k}),$$
$$s_0^2 = \varepsilon_*^2 \sum_k (2L_{1,k} + L_{2,k})^2, \qquad s_f^2 = \varepsilon_*^2 \sum_k c_{1,k}^2,$$
$$b_f = \tfrac{\sqrt{n}}{2} + 1 + \tfrac{\varepsilon_*^2}{2}\sum_k c_{2,k} + \tfrac{\sqrt{n}+1}{\sqrt{2}}, \qquad k_2 = 2b_f^2 + (4 + 8b_{\mathrm{B}}^2)s_f^2.$$

Then $\ell_b = \kappa_n R + b_0$, $\ell_\sigma = s_0$, $B = b_f$, $S = s_f$ are admissible up to σ_R. Indeed, each vector of normalized weights has sum at most one. The position and normalized-weight components of the deterministic CH drift have respective bounds $R(\sqrt{n}\,|q - Q| + \frac{n}{2}|(M^c - P)/R|)$ and $R(\sqrt{n}\,|q - Q| + \frac{n+1}{2}|(M^c - P)/R|)$. Their sum is bounded by $\kappa_n R|z|$. We apply the mean value theorem to f_k, h_k, ξ_k and ξ_k' to bound the noise drift and diffusion. The normalized weights sum to at most one, so the respective bounds are $b_0|z|$ and $s_0|z|$, with b_0, s_0 as displayed above. The interactions among localized measures contribute a residual bounded by $(\sqrt{n}/2 + 1 + \varepsilon_*^2 \sum c_{2,k}/2)L$. The remainder adds at most $(\sqrt{n}+1)\sqrt{\mathcal{R}/2}$ to the normalized drift and no additional noise residual. In particular, none of these bounds contains the full mass of m.

For the profile estimate, we use the following constants. In addition to H, Λ in Theorem 6.1, write

$$\begin{aligned}
J^2 &= 2r + 2r^3/3, \quad B_L^2 = nJ^2, \quad c_{\mathcal{R}} = b_E + v_E/2,\\
c_L &= 2\sqrt{2} + \varepsilon_*^2 \sum_k (18L_{1,k}^2 + 2L_{0,k}L_{2,k}),\\
b_1 &= 4b_0 + 2 + (4 + 8b_{\mathrm{B}}^2)s_0^2,\\
Z_T &= e^{b_1 T}\left[2H^2 + 4k_2 T(e^{c_L T}B_L^2 + e^{c_{\mathcal{R}} T}\Lambda^2)\right],\\
C_T &= 4\sqrt{2}B_L e^{c_L T/2} + 4nZ_T + 8\sqrt{Z_T} + 4\Lambda^2 e^{c_{\mathcal{R}} T}.
\end{aligned} \tag{6.25}$$

The initial projection estimate gives $Y_0^{\mathrm{sc}} \le H^2\delta^2$. The bounds for L and $\mathcal{R}$ give

$$\mathbb{E}F_{t\wedge\sigma_R}^2 \le 4e^{2RT}(e^{c_L T}B_L^2 + e^{c_{\mathcal{R}} T}\Lambda^2)\delta^2.$$

Applying Gronwall's inequality to Lemma 6.9, we obtain

$$\mathbb{E}\sup_{t\le T} Y_{t\wedge\sigma_R}^{\mathrm{sc}} \le Z_T e^{\alpha_n RT}\delta^2.$$

We apply Lemma 6.6 to the localized measures and then add the remainder velocity w. The reconstruction argument uses only positivity and the bound on the sum of the component masses, so it applies with c, M^c, L under the common local stopping time σ_R. For the full velocity profile, we have

$$\|u - v\|_{H^1}^2 \le 4RL + 4nR^2Y^{\mathrm{sc}} + 8R^2\sqrt{Y^{\mathrm{sc}}} + 2\mathcal{R} \quad \text{up to } \sigma_R.$$

Since $R \geq 1$, $\delta \leq 1$ and $\alpha_n \geq 2$, the preceding estimates give the explicit stopped comparison

$$\mathbb{E} \sup_{t \leq T} \|u_{t\wedge\sigma_R} - v_{t\wedge\sigma_R}\|_{H^1}^2 \leq C_T R^2 e^{\alpha_n RT} \delta. \tag{6.26}$$

6.5 Exit probabilities and moments on the full time interval

Proof of Theorems 1.3 and 6.1 It remains to remove σ_R. The localized total momentum M^c changes through its deterministic interaction with the remainder. With $S^c = M^c + P$ its drift from that interaction is $-\int_{\mathbb{R}} \bar{w}_x \, \mathrm{d}m^c$. Before σ_R this has absolute value at most $M^c/\sqrt{2} \leq S^c/\sqrt{2}$. The relative noise coefficients satisfy the same bounds as for the full momentum, namely $\varepsilon^2 A_m$ and $\varepsilon^2 V_m$. Also $S_0^c \leq S_*$, and S^c is positive because the reference momentum is positive. Consequently

$$\log(S^c_{t\wedge\sigma_R}/S^c_0) \leq (2^{-1/2} + \varepsilon^2 A_m)T + N_{t\wedge\sigma_R}, \qquad [N]_{T\wedge\sigma_R} \leq \varepsilon^2 V_m T.$$

By the exponential martingale inequality, the probability that, by time T, the mass boundary is reached at σ_R is bounded by $\mathcal{B}_R(T)$. When $\varepsilon^2 V_m = 0$, the localized total momentum still exchanges momentum with the remainder. The deterministic upper bound is $S_* e^{T/\sqrt{2}}$, which gives the threshold in (6.3). Although the full momentum is conserved in this case, M^c need not be conserved.

If the other boundary is reached, $\mathcal{R}_{\sigma_R} = 1$ by continuity. The remainder energy estimate and $\mathcal{R}_0 \leq \Lambda^2 \delta^2$ imply

$$\mathbb{P}(\sigma_R \leq T) \leq \mathcal{B}_R(T) + \mathcal{D}_R(T)\Lambda^2\delta^2.$$

The union bound also covers simultaneous exits through both boundaries. Splitting the profile event at σ_R and using (6.26) proves (6.2).

To obtain moments without stopping, write $\mathcal{H}_t = \|u_t\|_{H^1}^2 + \|v_t\|_{H^1}^2$. For each full solution the deterministic energy drift vanishes. The same noise kernel bounds as in (6.1) show that $\mathrm{d}\mathcal{H} = \alpha\mathcal{H}\,\mathrm{d}t + \mathcal{H}\gamma \cdot \mathrm{d}W$, with $\alpha \leq b_E$ and $|\gamma|^2 \leq v_E$. Its scalar stochastic exponential gives, for every $s \geq 1$,

$$\mathbb{E} \sup_{t \leq T} \mathcal{H}_t^s \leq 2\mathcal{H}_0^s e^{[sb_E + (s^2 - s/2)v_E]T}.$$

Factor the s-th power into $\mathcal{E}(s\gamma \cdot W)$ and a drift exponential; the L^2 Doob inequality bounds the expected supremum of the stochastic exponential by $2e^{s^2 v_E T/2}$. Since $\delta \leq 1$, set $\mathcal{H}_* = (\|v_0\|_{H^1} + 1)^2 + \|v_0\|_{H^1}^2$. For $q \geq 2$ we have

$$\mathbb{E}\mathcal{X}_T^q \leq J_q := 2^{q/2+1}\mathcal{H}_*^{q/2} \exp\Big\{ \Big[\tfrac{q}{2} b_E + (\tfrac{q^2}{4} - \tfrac{q}{4}) v_E \Big] T \Big\}. \tag{6.27}$$

For any $q > 2$, applying Hölder's inequality on the exit event gives the following explicit second-moment estimate on $[0, T]$:

$$\mathbb{E}\mathcal{X}_T^2 \leq C_T R^2 e^{\alpha_n RT}\delta + J_q^{2/q} \min\{1, \mathcal{B}_R(T) + \mathcal{D}_R(T)\Lambda^2\delta^2\}^{1-2/q}. \tag{6.28}$$

This argument does not require independence of the exit event and the energies. For each fixed R, take the upper limit as $\delta \to 0$, and then let $R \to \infty$ to obtain L^2 convergence. Jensen's inequality gives convergence for every smaller moment. Interpolation with (6.27), using $q > p$, gives convergence for each finite $p > 2$. The reference and all constants are fixed along the initial sequence; the initial energy bound follows from its H^1 convergence. Thus these estimates apply to the entire sequence.

For an explicit modulus when $\varepsilon^2 V_m > 0$, fix $R_* > \max\{1, S_*\}$ and, for $0 < \delta < 1$, choose

$$R_\delta = R_* + \frac{\log(1/\delta)}{2\alpha_n T}.$$

The first term of (6.28) is $C_T e^{\alpha_n R_* T} R_\delta^2 \delta^{1/2}$, and the remainder's exit-probability contribution $\mathcal{D}_{R_\delta}(T)\Lambda^2\delta^2$ is $O(\delta^{2-1/\alpha_n})$. Given $c < (2\varepsilon^2 V_m T)^{-1}$, choose one fixed $q > 2/(1 - 2c\varepsilon^2 V_m T)$ and set $\alpha = 1-2/q$. Then $\alpha/(2\varepsilon^2 V_m T) > c$. Since $\log R_\delta = \log\log(1/\delta) - \log(2\alpha_n T) + o(1)$, the α-th power of the mass-exit bound is at most $C_c e^{-c[\log\log(1/\delta)]^2}$ for every $c < (2\varepsilon^2 V_m T)^{-1}$. The remainder contribution to the second moment is $O(\delta^{(2-1/\alpha_n)\alpha})$. Absorbing the terms that decay algebraically into this bound proves (6.4). For $\varepsilon^2 V_m = 0$, choose the fixed $R > \max\{1, S_* e^{T/\sqrt{2}}\}$ instead. The right-hand side of (6.28) with $q = 4$ is then $O(\delta)$. If $\delta = 0$, uniqueness from Theorem 2.2 gives identical solutions. □

Remark 6.10 (Scope of the estimate on a fixed time interval) For spatially constant noise the common random translation reduces the evolution to deterministic CH, and the exponent $1/2$ in the H^1 estimate agrees with the example in Section 7.1. For general noise, the displayed modulus on a fixed time interval comes from the Gaussian tail for the logarithm of $M^c + P$ and the exponential dependence of the stopped comparison on its threshold. The sharpness statement concerns the spatially constant case.

7 Local estimates for peakon weights and positions

The preceding section compares velocity profiles under arbitrary nonnegative momentum perturbations. When both profiles are finite positive multipeakons, their weights and positions can be recovered as in Theorem 1.4. The quantitative statement below specifies the threshold for identifying and matching the peaks. Throughout this section, $G(x) = \frac{1}{2}e^{-|x|}$ and the coefficients of G are momentum weights.

Theorem 7.1 (Explicit local estimates for weights and positions) *Let*

$$U = \sum_{i=1}^{n} p_i G(\cdot - q_i), \qquad q_1 < \cdots < q_n, \qquad p_i \geq p_* > 0, \qquad q_{i+1} - q_i \geq \ell > 0.$$

For $n = 1$, the gap condition is vacuous, and any $\ell > 0$ may be chosen. Let $V = \sum_{j=1}^{s} a_j G(\cdot - y_j)$ have $s \leq n$ distinct ordered nodes and positive weights; $s = 0$ is allowed. Set

$$\begin{gathered} r = \min\{\ell/4, 1\}, \qquad H_r = \frac{2}{r} + \frac{2r}{3}, \qquad A_r = \frac{2}{r} + \frac{8r}{3}, \qquad C_r = 2 + \frac{2r^2}{3} = rH_r, \\ \delta = \|U - V\|_{H^1}, \qquad \delta_* = \frac{p_*}{2\sqrt{A_r}}. \end{gathered} \tag{7.1}$$

If $\delta \leq \delta_*$, then $s = n$, and the ordered matching is the unique matching for which $y_i \in (q_i - r, q_i + r)$. Moreover, with $d_i = |y_i - q_i|$,

$$\sum_{i=1}^{n} |a_i - p_i|^2 \leq A_r \delta^2, \tag{7.2}$$

$$\sum_{i=1}^{n} \frac{p_i^2 d_i}{2 + 2d_i^2/3} \leq \delta^2, \qquad \sum_{i=1}^{n} p_i^2 d_i \leq C_r \delta^2. \tag{7.3}$$

In particular,

$$|a_i - p_i| \leq \sqrt{A_r}\delta, \qquad d_i \leq \frac{C_r}{p_i^2}\delta^2, \qquad \tfrac{1}{2}p_i \leq a_i \leq \tfrac{3}{2}p_i, \qquad d_i < r/4.$$

The approximating gaps are at least $7\ell/8$. The constants require no upper bound on the weights and no bound on the total mass or reference diameter.

The reproducing identity

$$\langle G(\cdot - q), f\rangle_{H^1} = f(q), \qquad \langle G(\cdot - q), G(\cdot - y)\rangle_{H^1} = \tfrac{1}{2}e^{-|q-y|} \tag{7.4}$$

holds when $f \in H^1(\mathbb{R})$ is identified with its continuous representative. The identity follows from $(1 - \partial_x^2)G = \delta_0$ by density.

Proof of the local parameter estimates We prove Theorem 7.1, which implies Theorem 1.4. Write $e = U - V$. For $h > 0$, the triangular hat $T_{q,h}(x) = (1 - |x - q|/h)_+$ satisfies

$$\|T_{q,h}\|_{H^1}^2 = \frac{2}{h} + \frac{2h}{3}. \tag{7.5}$$

The intervals $I_i = (q_i - r, q_i + r)$ are pairwise disjoint. If I_i had no approximating node, then $T_{q_i,r}$ would vanish at every y_j, including nodes at the endpoints. Its support contains no other reference node. Hence (7.4) gives

$$p_i = \langle e, T_{q_i,r}\rangle_{H^1} \le \delta\sqrt{H_r} \le \frac{p_*\sqrt{H_r}}{2\sqrt{A_r}} < p_*,$$

a contradiction. Each of the n intervals therefore contains an approximating node. Since $s \le n$, every interval contains exactly one node, no node lies outside their union, and $s = n$. The ordering then determines the matching uniquely.

Define the plateau hat P_i to equal 1 on $[q_i - r, q_i + r]$, to decrease linearly to zero on each adjacent interval of length r, and to vanish outside $[q_i - 2r, q_i + 2r]$. Integration gives

$$\|P_i\|_{H^1}^2 = \frac{2}{r} + \frac{8r}{3} = A_r.$$

Since $4r \le \ell$, these supports have disjoint interiors, and the $P_i/\sqrt{A_r}$ are orthonormal in H^1. Every other reference node lies outside the support of P_i. For $j \ne i$, $|y_j - q_i| \ge \ell - r \ge 3r$, so every other approximating node also lies outside that support. The matched node lies on the plateau of the hat. Thus $\langle e, P_i\rangle_{H^1} = p_i - a_i$, and Bessel's inequality proves (7.2).

For each i with $d_i > 0$, take the smaller hat T_{q_i,d_i}. The matched node lies at an endpoint where the hat vanishes. Every other approximating node is outside its support, and q_i is the only reference node in its support. Consequently, $\langle e, T_{q_i,d_i}\rangle_{H^1} = p_i$. The supports of these hats also have disjoint interiors. Bessel's inequality and (7.5) yield

$$\sum_{i:d_i>0} \frac{p_i^2}{2/d_i + 2d_i/3} \le \|e\|_{H^1}^2 = \delta^2.$$

The indices with $d_i = 0$ contribute zero, so no test is needed for them. Since $d_i < r$, the preceding inequality proves both estimates in (7.3). The weight ratios follow from $\sqrt{A_r}\delta \le p_*/2 \le p_i/2$, while

$$d_i \le \frac{C_r\delta_*^2}{p_*^2} = \frac{rH_r}{4A_r} < \frac{r}{4}.$$

Finally $y_{i+1} - y_i \ge \ell - d_i - d_{i+1} \ge \ell - r/2 \ge 7\ell/8$. □

The counting argument requires the strict condition $\delta < p_*/\sqrt{H_r}$. The threshold in (7.1) leaves a strict margin, so the theorem permits $\delta \le \delta_*$. Both the reference and the approximating profiles belong to the positive-momentum class considered throughout the paper.

Corollary 7.2 (Comparison of profile and parameter distances) *Fix the reference weights $p = (p_1, \dots, p_n)$ and equip the ordered parameter coordinates with the metric*

$$d_p\big((a,y),(b,z)\big)^2 = \sum_i |a_i - b_i|^2 + \sum_i p_i^2 |y_i - z_i|.$$

For the matching in Theorem 7.1, put $D = d_p((a,y),(p,q))$. Then

$$\frac{2}{3n}\|V-U\|_{H^1}^2 \le D^2 \le (A_r + C_r)\|V-U\|_{H^1}^2.$$

If additionally $\sum_i p_i \le M$, then

$$\|V-U\|_{H^1} \le \frac{1}{\sqrt{2}} \sum_i |a_i - p_i| + M\sqrt{\max_i |y_i - q_i|}. \tag{7.6}$$

Proof The upper bound for D^2 follows by adding (7.2) and (7.3). To prove the reverse bound, write

$$V - U = \sum_i (a_i - p_i) G(\cdot - y_i) + \sum_i p_i \big(G(\cdot - y_i) - G(\cdot - q_i)\big).$$

By (7.4), $\|G(\cdot - y) - G(\cdot - q)\|_{H^1}^2 = 1 - e^{-|y-q|} \le |y - q|$. The triangle and Cauchy–Schwarz inequalities therefore give

$$\|V-U\|_{H^1} \le \sqrt{n/2}\Big(\sum_i |a_i - p_i|^2\Big)^{1/2} + \sqrt{n}\Big(\sum_i p_i^2 |y_i - q_i|\Big)^{1/2} \le \sqrt{3n/2}\, D.$$

Applying the triangle inequality to the same decomposition gives (7.6). □

Corollary 7.3 (Parameter estimates along a reference solution) *Let I be any time interval, possibly a stopped interval on a fixed sample path. Suppose $U_t = \sum_{i=1}^n p_i(t) G(\cdot - q_i(t))$ has ordered parameters with $p_i(t) \ge p_*$ and gaps at least ℓ, uniformly on I. Let $u_t \in H^1(\mathbb{R})$ and choose any positive multipeakon profile V_t with at most n peaks satisfying*

$$\|V_t - u_t\|_{H^1} \le \eta_t, \qquad \|u_t - U_t\|_{H^1} \le \rho_t, \qquad \eta_t + \rho_t \le \delta_*.$$

Then every V_t has exactly n peaks, and its ordered parameters obey

$$\|a(t) - p(t)\|_{\ell^2} \le \sqrt{A_r}\,(\eta_t + \rho_t),$$
$$\sum_i p_i(t)^2 |y_i(t) - q_i(t)| \le C_r(\eta_t + \rho_t)^2.$$

In particular, if $\Delta = \sup_{t\in I}(\eta_t + \rho_t) \le \delta_*$, the corresponding supremum bounds are $\sqrt{A_r}\Delta$ for the weight ℓ^2 error and $C_r\Delta^2/p_*^2$ for the largest position error.

Proof The triangle inequality gives $\|V_t - U_t\|_{H^1} \le \eta_t + \rho_t$. Apply Theorem 7.1 at each time. □

This pathwise statement requires neither adaptedness nor continuity of the chosen approximant, and asserts neither property for a selection. Probability statements additionally require measurability of the errors or events involved. The bound on ρ_t is indispensable: distance to the full peakon family alone does not give proximity to a prescribed reference trajectory. For example, $pG(\cdot - L)$ has zero one-peakon family distance for every L, although its displacement from pG is arbitrary.

Sharpness and necessary hypotheses.
For a single peak, the exact formula is

$$\|aG(\cdot - y) - pG(\cdot - q)\|_{H^1}^2 = \tfrac{1}{2}(a-p)^2 + ap(1 - e^{-|y-q|}).$$

With $a = p$ and error $\delta < p$, this gives $|y - q| = -\log(1 - \delta^2/p^2) = \delta^2/p^2 + O(\delta^4)$; with $y = q$, it gives $|a - p| = \sqrt{2}\delta$. Thus the location exponent 2 and weight exponent 1 cannot be improved.

The following examples explain the distinct roles of the lower weight bound, the separation condition and the bound on the number of peaks. For fixed $L > 1$, the profiles $G + \varepsilon G(\cdot - L)$ and $G + \varepsilon G(\cdot - (L+1))$ have position error 1 but squared velocity error $\varepsilon^2(1 - e^{-1})$. For a fixed $0 < c < 1$, the profiles $G(\cdot + h) + G(\cdot - h)$ and $(1+c)G(\cdot + h) + (1-c)G(\cdot - h)$ have weight errors c but squared velocity error $c^2(1 - e^{-2h})$. Moreover,

$$\|G(\cdot + h) + G(\cdot - h) - 2G\|_{H^1}^2 = 3 + e^{-2h} - 4e^{-h} = 2h + O(h^3).$$

Thus no uniform counting threshold persists as the reference gaps collapse. Without the upper bound on the number of peaks, a single peak can be split into two nearby half-weight peaks, or supplemented by $\varepsilon G(\cdot - L)$ at arbitrary distance with velocity error $\varepsilon/\sqrt{2}$. All these examples have bounded total mass. Upper weight bounds are unnecessary for the inverse theorem but are needed for an unweighted forward bound such as (7.6): the displacement p^{-2} between pG and $pG(\cdot - p^{-2})$ tends to zero as $p \to \infty$, while their squared velocity distance tends to 1.

7.1 The sharp exponent for spatially constant noise

Proposition 7.4 (The one-half exponent is necessary) *Fix $c > 0$ and $T > 0$. In the admissible subclass of constant noise fields, no uniform estimate of the form $\|u_T - v_T\|_{H^1} \le C(\|u_0 - v_0\|_{H^1} + |M_0 - P_0|)^\alpha$ can hold near a single reference peakon for $\alpha > 1/2$. The same obstruction applies to bounds for expected squared error with initial power greater than one. Thus the one-half exponent in the main theorem is optimal over the class $n \ge 1$; the example below already occurs for $n = 1$.*

Proof For constant ξ_k, both solutions are translated by the common $B_t = \varepsilon \sum_k \xi_k W_t^k$. Take

$$v_t = ce^{-|\cdot - ct - B_t|}, \qquad u_t = (c+h)e^{-|\cdot - (c+h)t - B_t|}, \quad h > 0.$$

Their initial H^1 and momentum errors are $\sqrt{2}h$ and $2h$. The Green inner product gives exactly

$$\|u_t - v_t\|_{H^1}^2 = 2h^2 + 4c(c+h)(1 - e^{-ht}) = 4c^2 th + O(h^2) \quad (h \downarrow 0).$$

At $t = T > 0$, the coefficient of h is positive. The distance is independent of the common translation, so the same computation also rules out the stated moment estimate and proves the assertion. □

8 Discussion

The two stability statements address different comparisons in the same energy topology. The spectral defect controls distance to the multipeakon set without fixing peak weights or gaps, and yields exit estimates on noise-dependent time intervals. The localized momentum decomposition controls the full H^1 path error relative to one specified multipeakon driven by the same Brownian motions, on each fixed finite interval. Both comparisons allow perturbations whose total momenta have no uniform bound.

For finite positive multipeakon profiles, the geometric estimate identifies individual peaks near a reference with distinct positions and positive weights. Its linear weight errors and quadratic position errors express how the cusp of the Green kernel converts an energy error into parameter errors. This interpretation applies to finite profiles, while the path comparison also admits diffuse momentum.

A Approximation with prescribed total momentum

In this appendix, the approximating multipeakon is required to have the same total momentum as m. A Gram determinant identity yields a nearest-center estimate, which in turn gives the approximation inequality and exit bounds for the distance $d_{n,M}$ defined below. We use the elementary symmetric functions e_r from Section 3, with $e_2 = M^2/2 - E = D$.

A.1 Gram determinants and positive atomic approximation

The elementary symmetric functions defined above have the Gram determinant representation

$$e_r(m) = \frac{1}{r!} \int_{\mathbb{R}^r} \det[e^{-|x_i - x_\ell|/2}]_{i,\ell=1}^r \, \mathrm{d}m^{\otimes r}. \tag{A.1}$$

For ordered points $x_1 \le \cdots \le x_r$, elementary row operations give the kernel formula

$$\det[e^{-|x_i - x_\ell|/2}] = \prod_{i=1}^{r-1} (1 - e^{-(x_{i+1} - x_i)}). \tag{A.2}$$

It vanishes at repeated points. To justify (A.1) for a general finite measure, let P_d be the projections onto the first d vectors of an orthonormal basis of $\mathcal{H}$. The finite-dimensional Gram identity applied to $P_d \mathcal{C}_m P_d$ gives

$$e_r(P_d \mathcal{C}_m P_d) = \frac{1}{r!} \int_{\mathbb{R}^r} \det[\langle P_d k_{x_i}, P_d k_{x_\ell} \rangle_{\mathcal{H}}] \, \mathrm{d}m^{\otimes r}.$$

The determinants converge pointwise to the kernel in (A.1); the Gram determinant inequality bounds them between zero and one, so dominated convergence applies. On the operator side, $P_d \mathcal{C}_m P_d \to \mathcal{C}_m$ in trace norm; for example,

$$\|\mathcal{C}_m - P_d \mathcal{C}_m P_d\|_1 \le 2 \|\mathcal{C}_m^{1/2}\|_{\mathrm{HS}} \|(I - P_d)\mathcal{C}_m^{1/2}\|_{\mathrm{HS}} \longrightarrow 0.$$

Newton's identities and continuity of traces of fixed powers in trace norm therefore give convergence of e_r as well. In particular,

$$e_1 = M, \qquad e_2 = D, \qquad e_3 = \frac{M^3 - 3Mp_2 + 2p_3}{6} = \frac{M^3}{6} - ME + 2\mathcal{F}.$$

If the support contains r distinct points, choose disjoint neighborhoods of them, each with positive measure. The determinant in (A.2) is strictly positive on their product, so its integral is positive. Conversely, if the support has at most $r-1$ points, every tuple has a repeated point. Thus $e_r = 0$ exactly when the measure has at most $r-1$ support points.

For the additional estimates with prescribed total momentum, set

$$\mathcal{P}_{n,M} = \left\{ \sum_{a=1}^{s} p_a G(\cdot - q_a) : 1 \le s \le n,\ p_a > 0,\ \sum_a p_a = M \right\},$$
$$d_{n,M}(m) = \operatorname{dist}_{H^1}(u_m, \mathcal{P}_{n,M(m)}), \qquad M > 0,$$

and set $d_{n,M}(0) = 0$. The unrestricted family in the introduction satisfies $\mathcal{P}_n = \bigcup_{M>0} \mathcal{P}_{n,M}$.

Proposition A.1 (Approximation with prescribed total momentum) *For every finite nonnegative measure and $n \ge 1$,*

$$d_{n,M}(m)^{2n} \le a_n M^{n-1} e_{n+1}(m), \qquad a_n = 2^{n-2}(n+1)!. \tag{A.3}$$

In particular $a_1 = 1$ and $d_{2,M}^4 \le 6Me_3$. When $e_n > 0$, there is also the estimate

$$d_{n,M}^2 \le 2(n+1)M \frac{e_{n+1}}{e_n}. \tag{A.4}$$

Proof The zero measure satisfies the assertion by convention. Assume henceforth that $M > 0$, and write $K(x,y) = 1 - e^{-|x-y|}$ and

$$q_n(m) = \inf_{z_1,\dots,z_n} \int_{\mathbb{R}} \min_a K(x, z_a) m(\,\mathrm{d}x).$$

For a fixed set of distinct ordered centers $Q = (z_1, \dots, z_n)$, let $\Delta(Q)$ be its Gram determinant and $R_Q(x) = \Delta(Q \cup \{x\})/\Delta(Q)$. The product formula shows that outside the extreme centers $R_Q(x) = K(x, z_{\text{nearest}})$, while for $z_a < x < z_{a+1}$,

$$R_Q(x) = \frac{K(x,z_a)K(x,z_{a+1})}{K(z_a,z_{a+1})}.$$

Since $K(z_a, z_{a+1}) \le K(x,z_a) + K(x,z_{a+1})$,

$$\min_a K(x, z_a) \le 2R_Q(x).$$

Multiply by $\Delta(Q)$ and extend the resulting inequality symmetrically to all tuples $Q \in \mathbb{R}^n$. At repeated centers, both sides vanish, so the multiplied expression requires no division there. Integrate over $\mathbb{R}^n \times \mathbb{R}$ against $m^{\otimes n}(\,\mathrm{d}Q)m(\,\mathrm{d}x)$. The inner integral on the left is at least $q_n(m)$, while (A.1) gives the factors $n!$ and $(n+1)!$. Consequently,

$$q_n(m)\, n! e_n(m) \le 2(n+1)! e_{n+1}(m).$$

For fixed centers, choose Borel nearest-center cells and transfer the mass of each cell to its center. This gives a positive atomic measure of mass M on at most n points. As $\|G(\cdot - x) - G(\cdot - y)\|_{H^1}^2 = K(x, y)$, the Bochner Cauchy–Schwarz inequality gives

$$\left\| G * m - \sum_a m(\mathrm{cell}_a) G(\cdot - z_a) \right\|_{H^1}^2 \le M \int_{\mathbb{R}} \min_a K(x, z_a) m(\,\mathrm{d}x).$$

Taking infima proves (A.4), without requiring a minimizing set of centers. If $e_n = 0$, the measure is already supported on at most $n - 1$ points and (A.3) is immediate. The ratio estimate is asserted only for $e_n > 0$.

For $n = 1$, expand

$$\|u_m - MG(\cdot - y)\|_{H^1}^2 = E + M^2/2 - 2Mu_m(y).$$

Averaging against $m(\,\mathrm{d}y)/M$ gives $M^2/2 - E = e_2$. The infimum over y therefore satisfies $d_{1,M}^2 \le e_2$, which is (A.3) with $a_1 = 1$. For $n \ge 2$, apply the induction hypothesis and $d_{n,M} \le d_{n-1,M}$, then multiply by (A.4). The constants obey $a_n = 2(n+1)a_{n-1}$, which gives the formula. □

A.2 Stochastic estimates and exit times

Newton's identities express each e_r as a polynomial in $p_1, \dots, p_r$. By Lemma 3.2 and the scalar Itô formula, each has an exact weak stochastic balance, with coefficients given by the first and second derivatives along the deterministic coadjoint noise flow. The product formula (A.2) also gives direct bounds. For one ξ, write $L_j = \|\xi^{(j)}\|_\infty$. On an ordered tuple, let $g_i = x_{i+1} - x_i > 0$. The logarithmic derivative of the determinant together with its r momentum weights is

$$c = \sum_{i=1}^{r-1} \frac{\xi(x_{i+1}) - \xi(x_i)}{e^{g_i} - 1} - \sum_{i=1}^{r} \xi'(x_i).$$

Thus $|c| \le (2r-1)L_1$. Let $V = \sum_i \xi(x_i)\partial_i$. Writing $d_i = \xi(x_{i+1}) - \xi(x_i)$ and $w_i = (\xi\xi')(x_{i+1}) - (\xi\xi')(x_i)$, one has

$$V\left(\frac{d_i}{e^{g_i} - 1}\right) = \frac{w_i}{e^{g_i} - 1} - \frac{d_i^2 e^{g_i}}{(e^{g_i} - 1)^2}.$$

Using $g/(e^g - 1) \le 1$, $g^2 e^g/(e^g - 1)^2 \le 1$, $|d_i| \le L_1 g_i$, and $|w_i| \le (L_1^2 + L_0 L_2) g_i$ gives the following bounds, with the differentiated determinant extended by zero at repeated points:

$$\begin{aligned} |e_r'| &\le (2r-1)L_1 e_r, \\ |e_r''| &\le \left(\left[(2r-1)^2 + 2(r-1) \right] L_1^2 + (2r-1)L_0 L_2 \right) e_r. \end{aligned} \tag{A.5}$$

The formula remains valid for $r = 1$, with empty gap sums.

Set

$$Z_n = M^{n-1} e_{n+1}, \quad \ell_n = \frac{\varepsilon^2}{2} \sum_k \left((9n^2 + 2n) L_{1,k}^2 + 3n L_{0,k} L_{2,k} \right), \quad v_n = 9n^2 \sum_k L_{1,k}^2.$$

The total momentum satisfies $|M_k'| \le L_{1,k} M$ and $|M_k''| \le (L_{1,k}^2 + L_{0,k} L_{2,k}) M$. Applying the product rule and using (A.5), we obtain

$$|Z_{n,k}'| \le 3n L_{1,k} Z_n, \qquad |Z_{n,k}''| \le \left((9n^2 + 2n) L_{1,k}^2 + 3n L_{0,k} L_{2,k}\right) Z_n. \tag{A.6}$$

Theorem A.2 (Stability with prescribed total momentum) *Fix an integer $n \ge 1$, a threshold $\eta > 0$ and $0 \le T < \infty$. The process $e^{-\ell_n t} Z_n(m_t)$ is a nonnegative continuous supermartingale. Put*

$$\tau_{n,\eta} = \inf\{t : d_{n,M}(m_t) \ge \eta\}, \qquad b_{n,\eta} = \eta^{2n}/a_n.$$

Then

$$\mathbb{P}(\tau_{n,\eta} \le T) \le \min\{1, e^{\ell_n T} Z_n(m_0)/b_{n,\eta}\}. \tag{A.7}$$

If $Z_n(m_0) > 0$ and $\varepsilon^2 v_n T > 0$, the logarithmic bound is

$$\mathbb{P}(\tau_{n,\eta} \le T) \le \exp\left\{ -\frac{[\log(b_{n,\eta}/Z_n(m_0)) - \ell_n T]_+^2}{2\varepsilon^2 v_n T} \right\}.$$

If $0 < Z_n(m_0) < b_{n,\eta}$ and $\ell_n > 0$, then

$$\mathbb{E}\tau_{n,\eta} \ge \ell_n^{-1} \log \frac{b_{n,\eta}}{Z_n(m_0)}.$$

The same bounds apply to the distance to the family with at most n peaks and arbitrary total momentum. Initial support on at most n points is invariant in this class.

Proof The exact scalar balance and (A.6) yield the supermartingale property. To verify continuity of the distance, consider two measures and rescale any competing atomic profile to obtain

$$|d_{n,M}(m) - d_{n,M}(\widetilde{m})| \le \|G * m - G * \widetilde{m}\|_{H^1} + |M(m) - M(\widetilde{m})|/\sqrt{2}.$$

This bound also holds when a total momentum is zero. An exit requires $Z_n \ge b_{n,\eta}$ by Proposition A.1. Apply Lemma 5.1 with $Y = Z_n$, $a = \ell_n$, $v = \varepsilon^2 v_n$ and $b = b_{n,\eta}$. This gives the probability bound, the logarithmic bound and the mean exit-time bound, including the zero initial value case. The support characterization following (A.2) proves invariance. Constant coefficients or $\varepsilon = 0$ give $\ell_n = 0$ and a conserved functional Z_n. In this case $\tau_{n,\eta} = \infty$ if $Z_n(m_0) < b_{n,\eta}$; no division by a zero constant is used. □

For two peaks the formulas reduce to

$$Z_2 = M\left(\frac{M^3}{6} - ME + 2\mathcal{F}\right), \quad d_{2,M}^4 \le 6 Z_2, \quad \ell_2 = \frac{\varepsilon^2}{2} \sum_k (40 L_{1,k}^2 + 6 L_{0,k} L_{2,k}).$$

This estimate controls distance to the family with at most two peaks. It does not prescribe the two amplitudes or their separation, or compare the solution with a specified two-particle orbit.

A.3 Initial errors and normalized profiles

We next bound the initial value of Z_n explicitly in terms of the H^1 error and the total-momentum error. We use the fixed-space covariance identity (3.1).

Corollary A.3 (Initial error estimate) *Fix an integer $n \geq 1$. Let $\widetilde{m}_0$ have at most n atoms, let both initial total momenta be at most $M_* > 0$, and suppose*

$$\|G * m_0 - G * \widetilde{m}_0\|_{H^1} \leq \delta, \qquad |M(m_0) - M(\widetilde{m}_0)| \leq \rho.$$

Then in (A.7) *one may use*

$$Z_n(m_0) \leq M_*^{2n-1} \left(\frac{\rho}{n!} + \sqrt{2}\,\delta \sum_{\ell=0}^{n-1} \frac{1}{\ell!} \right).$$

The constants are independent of the reference positions.

Proof Interpolate linearly between the two positive measures and their fixed-space operators. If C is an interpolated operator and $H = \mathcal{C}_{m_0} - \mathcal{C}_{\widetilde{m}_0}$, the Newton derivative formula is

$$De_{n+1}(C)[H] = \sum_{j=0}^{n} (-1)^j e_{n-j}(C) \operatorname{tr}(C^j H).$$

For $j = 0$, we use $|\operatorname{tr} H| \leq \rho$; for $j \geq 1$, we use $\|C^j\|_{\mathrm{HS}} \leq M_*^j$ and $\|H\|_{\mathrm{HS}} \leq \sqrt{2}\delta$. Also $e_{n-j}(C) \leq M_*^{n-j}/(n-j)!$. Integrating the derivative and using $e_{n+1}(\widetilde{m}_0) = 0$ proves the claim. □

To obtain an estimate invariant under momentum rescaling, assume $M(m_0) > 0$, so total momentum stays positive, and define

$$R_n = \frac{e_{n+1}}{M^{n+1}}, \qquad r_n(m) = \frac{d_{n,M}(m)}{M},$$
$$\widehat{\ell}_n = \frac{\varepsilon^2}{2} \sum_k \left((9n^2 + 16n + 6) L_{1,k}^2 + (3n+2) L_{0,k} L_{2,k} \right).$$

The total-momentum equation has bounded relative drift and diffusion coefficients. The stochastic exponential argument in Lemma 5.1 gives $M_t > 0$ on every finite interval. The reconstruction inequality gives $r_n^{2n} \leq a_n R_n$. Apply Itô's formula first up to the stopping times at which M leaves $[L^{-1}, L]$, and differentiate $M^{-(n+1)} e_{n+1}$ along each noise direction. Using (A.5), the resulting bounds are

$$|R'_{n,k}| \leq (3n+2) L_{1,k} R_n, \qquad |R''_{n,k}| \leq \left((9n^2 + 16n + 6) L_{1,k}^2 + (3n+2) L_{0,k} L_{2,k} \right) R_n.$$

These statements also cover $\varepsilon = 0$ without division by ε. The stopping times tend to infinity almost surely, because M is continuous and strictly positive on compact time intervals. Thus $e^{-\widehat{\ell}_n t} R_n(m_t)$ is a nonnegative local supermartingale and hence a supermartingale. In fact, $0 \leq R_n \leq 1/(n+1)!$ follows by expanding $(\sum_i \lambda_i)^{n+1}$.

For the initial estimate, use the static bound of Corollary A.3 on the normalized measure $m_0/M(m_0)$, with $M_* = 1$ and $\rho = 0$, and take the infimum over atomic measures of unit total momentum. This gives

$$R_n(m_0) \le \sqrt{2}\left(\sum_{\ell=0}^{n-1}\frac{1}{\ell!}\right) r_n(m_0).$$

Only the initial measure is normalized in this argument; the stochastic balance above continues to apply to the original solution. Applying the maximal inequality to the discounted supermartingale and using $r_n^{2n} \le a_n R_n$, we obtain, for $n \ge 1$, $\eta > 0$ and $0 \le T < \infty$,

$$\mathbb{P}\left(\sup_{t\le T} r_n(m_t) \ge \eta\right) \le \min\left\{1,\, a_n\sqrt{2}\left(\sum_{\ell=0}^{n-1}\frac{1}{\ell!}\right) e^{\widehat{\ell}_n T}\,\frac{r_n(m_0)}{\eta^{2n}}\right\}.$$

This normalized statement has no initial total-momentum bound. It is distinct from stability in the unnormalized H^1 distance using only an initial H^1 error.

B An auxiliary comparison using total momentum

The exhaustive partition in Section 6.2 provides a comparison of all component weights and positions when both the initial profile error and the total-momentum error are controlled. This additional initial information allows the comparison to be stopped only on full mass; the main theorem in Section 6 uses the local decomposition to remove that requirement.

$$\mu = |M_0 - P_0|, \qquad \delta_{\mathrm{tot}} = \delta + \mu.$$

For $T > 0$ and $R > M_0 + P_0$, recall $\theta_R = \inf\{t \ge 0 : M_t + P_t \ge R\}$ and define

$$\beta_R(T) = \begin{cases} \exp\left[-\dfrac{[\log(R/(M_0+P_0)) - \varepsilon^2 A_m T]_+^2}{2\varepsilon^2 V_m T}\right], & \varepsilon^2 V_m > 0, \\ 0, & \varepsilon^2 V_m = 0. \end{cases}$$

Theorem B.1 (Comparison using profile and total momentum errors) *Fix $T > 0$ and $R > M_0 + P_0$. Let m be a solution of Definition 2.1 with deterministic nonnegative finite initial momentum, driven by the same Brownian motions as the reference v. If $\delta_{\mathrm{tot}} \le 1$, there is a finite, explicit constant C_{tr} such that*

$$\mathbb{E}\sup_{t\le T}\|u_{t\wedge\theta_R} - v_{t\wedge\theta_R}\|_{H^1}^2 \le C_{\mathrm{tr}}\delta_{\mathrm{tot}}. \tag{B.1}$$

The constant depends only on n, R, T, ε_*, the C_b^3 noise norms, and the initial minimum reference gap. For $n = 1$ the gap dependence is absent. It does not depend on the absolute

reference positions, their diameter, a lower reference-weight bound, the number of atoms of m_0, or a spatial moment. No evolving-gap stopping time is used. For every $\eta > 0$,

$$\mathbb{P}\left(\sup_{t\le T}\|u_t - v_t\|_{H^1} \ge \eta\right) \le \min\left\{1,\ \beta_R(T) + \frac{C_{\mathrm{tr}}\delta_{\mathrm{tot}}}{\eta^2}\right\}. \tag{B.2}$$

For a fixed reference, convergence of the initial H^1 and total-momentum errors therefore implies convergence in probability in $C([0,T];H^1)$ to the specified reference solution driven by the same Brownian motions.

The constants in the SDE comparison below make the parameter dependence in the theorem explicit. Set

$$\begin{aligned} L_{\mathrm{CH}} &= \sqrt{n}(R+R^2) + \tfrac{1}{2}(n + R(n+1)),\\ L_b &= L_{\mathrm{CH}} + \tfrac{\varepsilon_*^2}{2}\sum_k (2H_{0,k} + RH_{1,k}),\\ L_\sigma^2 &= \varepsilon_*^2 \sum_k (2L_{1,k} + RL_{2,k})^2,\\ B &= \tfrac{\sqrt{n}}{2} + R + \tfrac{\varepsilon_*^2}{2}\sum_k c_{2,k}, \qquad S^2 = \varepsilon_*^2 \sum_k c_{1,k}^2. \end{aligned}$$

Define

$$K_1 = 6TL_b^2 + 24L_\sigma^2, \quad K_2 = 6TB^2 + 24S^2, \quad C_Z = e^{K_1 T}(3 + K_2 T C_L).$$

Here K_1, K_2 are constants, not kernels or noise fields.

Proposition B.2 (Comparison of group parameters) *Put*

$$A_0 = L_0^2 + \sum_i |M_i(0) - P_i^0|^2, \qquad Y_t = \sum_i (|q_i(t) - Q_i(t)|^2 + |M_i(t) - P_i(t)|^2).$$

Then

$$\mathbb{E}\sup_{t\le T} Y_{t\wedge\theta_R} \le C_Z A_0.$$

Proof On the ordered cone, the sign in V_i is the fixed sign of $i - j$. For nonnegative weight vectors with totals at most R, write $a = |q - Q|_2$ and $b = |M. - P.|_2$. The kernel inequalities give

$$\begin{aligned} |U(q,M.) - U(Q,P.)|_2, |V(q,M.) - V(Q,P.)|_2 &\le \sqrt{n}Ra + \tfrac{n}{2}b,\\ |M.V(q,M.) - P.V(Q,P.)|_2 &\le \sqrt{n}R^2 a + \tfrac{R(n+1)}{2}b. \end{aligned}$$

The $i = j$ term of V_i is identically zero. The mean value bounds for f_k, h_k, ξ_k, ξ_k' then show that the reference Itô drift and diffusion have Lipschitz constants L_b and L_σ in the Euclidean state $(q, M.)$. Formula (6.13) bounds the additional drift by BL and the Hilbert–Schmidt norm of the additional diffusion by SL. These estimates contain no inverse gap and remain valid as ordered configurations approach the boundary of the cone.

Subtract the reference equations from (6.12), and let $Z(t) = \mathbb{E}\sup_{s\le t} Y_{s\wedge\theta_R}$. The integral equation, Cauchy–Schwarz in time, and the L^2 Doob inequality give

$$Z(t) \le 3Y_0 + K_1\int_0^t Z(s)\,\mathrm{d}s + K_2\int_0^t \mathbb{E}L^2_{s\wedge\theta_R}\,\mathrm{d}s.$$

For example, the drift difference contributes

$$6TL_b^2\int_0^t Z(s)\,\mathrm{d}s + 6TB^2\int_0^t \mathbb{E}L^2_{s\wedge\theta_R}\,\mathrm{d}s,$$

and the martingale difference contributes

$$24L_\sigma^2\int_0^t Z(s)\,\mathrm{d}s + 24S^2\int_0^t \mathbb{E}L^2_{s\wedge\theta_R}\,\mathrm{d}s.$$

The stopped integrals contain the indicator $1_{\{s<\theta_R\}}$. Replacing their integrands by stopped suprema gives an upper bound. Since $q_i(0) = Q_i(0)$ for every i, Y_0 is the weight error in A_0. Applying Gronwall's inequality and using (6.11), we obtain the assertion. □

In particular, if $\mathscr{E}_R$ denotes the left side of (B.1), then

$$\mathscr{E}_R \le 2R\sqrt{C_L}L_0 + 2nC_ZA_0 + 4R^2\sqrt{C_ZA_0}. \tag{B.3}$$

This estimate holds without the smallness assumption $\delta_{\mathrm{tot}} \le 1$.

Lemma B.3 (Initial profile and momentum errors) *For $n \ge 2$, let $r_0 = \min\{\frac{1}{2}\min_i(Q^0_{i+1} - Q^0_i), 1\}$; for $n = 1$, set $r_0 = 1$. Define*

$$H_0^2 = 2/r_0 + 2r_0/3, \quad \kappa_0 = 1 - e^{-r_0}, \quad C_{\mathrm{init}} = 2nH_0^2 + (1 + 2/\kappa_0^2)\max(1, 2n). \tag{B.4}$$

Then $L_0 \le \mu + \sqrt{2n}\delta$ and $A_0 \le C_{\mathrm{init}}(\delta + \mu)^2$.

Proof Let $f(x) = \max_i e^{-|x-Q_i^0|}$. This continuous function belongs to H^1; its weak derivative is that of the maximizing exponential almost everywhere, and $\|f\|^2_{H^1} \le 2n$. On I_i it equals $e^{-|x-Q_i^0|}$, including ties. Since $f(Q_i^0) = 1$,

$$L_0 = M_0 - P_0 - \langle u_0 - v_0, f\rangle_{H^1} \le \mu + \sqrt{2n}\delta.$$

The pairing is justified by the Green reproducing identity, which holds for every H^1 test function identified with its continuous representative.

The compact tents $h_i(x) = (1 - |x - Q_i^0|/r_0)_+$ lie within the closures of the corresponding Voronoi cells, vanish at common boundaries, have norm H_0, and satisfy $h_i(Q_j^0) = 1_{i=j}$. For $d \ge 0$, $\min(d/r_0, 1) \le (1 - e^{-d})/\kappa_0$. Consequently

$$0 \le M_i(0) - \int_{\mathbb{R}} h_i\,\mathrm{d}m_0 \le L_i(0)/\kappa_0, \qquad |M_\cdot(0) - P^0_\cdot|_2 \le \sqrt{n}H_0\delta + L_0/\kappa_0.$$

Squaring and using the bound for L_0 gives (B.4). This argument requires neither a diameter bound nor a spatial moment assumption. □

Proof of Theorem B.1 For $\delta_{\mathrm{tot}} \le 1$, insert Lemma B.3 into (B.3). An admissible choice is

$$C_{\mathrm{tr}} = 2R\sqrt{2nC_L} + 2nC_Z C_{\mathrm{init}} + 4R^2\sqrt{C_Z C_{\mathrm{init}}}.$$

To control the mass exit, let $S_t = M_t + P_t$. The deterministic CH drift vanishes in both total-momentum equations. Positivity gives

$$\mathrm{d}S_t = a_t S_t\,\mathrm{d}t + S_t\gamma_t\cdot\,\mathrm{d}W_t, \quad |a_t| \le \varepsilon^2 A_m, \quad |\gamma_t|^2 \le \varepsilon^2 V_m.$$

Here the ratios are averages against the sum of the two positive momentum measures. Since $S_0 > 0$, its scalar stochastic exponential stays positive. Itô's formula implies

$$\log(S_t/S_0) \le \varepsilon^2 A_m t + N_t, \qquad [N]_t \le \varepsilon^2 V_m t.$$

The exponential martingale inequality for $\sup_{t\le T} N_t$ gives $\mathbb{P}(\theta_R \le T) \le \beta_R(T)$. If $\varepsilon^2 V_m = 0$, all effective noise derivatives vanish, so both momenta are constant and this exit probability is zero. Split the velocity event at θ_R and apply Markov's inequality to (B.1). This proves (B.2).

For this auxiliary comparison, convergence of the initial total momentum keeps S_0 bounded. With T fixed, first choose R large and then let δ_{tot} tend to zero while keeping R fixed. Theorem 1.3 removes the initial momentum requirement through the localized measure decomposition. □

B.1 Finite-particle initial data

Corollary B.4 (Additional initial information for finite particles) *Fix the specified reference initial profile. Suppose the perturbed initial momentum has at most n atoms and put $\delta = \|u_0 - v_0\|_{H^1}$. With the constants of Theorem 7.1, assume $\delta \le \min\{\delta_*, (1+\sqrt{nA_r})^{-1}\}$. The total-momentum error of the perturbation is bounded by $\sqrt{nA_r}\delta$, where A_r is the constant in Theorem 7.1 for the initial reference. Hence Theorem B.1 applies with $\delta_{\mathrm{tot}} := \delta + |M_0 - P_0| \le (1+\sqrt{nA_r})\delta$.*

Proof Theorem 7.1 gives exactly n matched atoms and an ℓ^2 weight error at most $\sqrt{A_r}\delta$. Summing the weight differences and applying Cauchy–Schwarz then yields the total-momentum estimate. The argument applies to profiles with at most n peaks; arbitrary nonnegative finite momentum perturbations may have large total mass at small H^1 error. □

C Relative derivatives of the Green operator

For completeness, we give the full proof of Lemma 3.1; see also Part I [22, Lemma 5.5].

Proof For real $f \in L^2(m)$, the signed measure $f\,m$ has finite total variation. Put

$$n_s = (\phi_s)_{\#}(J_s^{-1/2} f\,m), \qquad v_s = (-\partial_x^2 + 1/4)^{-1} n_s, \qquad Q_s = \langle f, T_s f\rangle = \|v_s\|_{\mathcal{H}}^2.$$

The signed measure evolves in the distributional sense according to $\partial_s n_s = -\xi\partial_x n_s - \frac{3}{2}\xi' n_s$. For smooth densities, integration by parts yields

$$\begin{aligned} Q_s' &= 2\int_{\mathbb{R}} (\xi v_{s,x} - \tfrac{1}{2}\xi' v_s)(v_s/4 - v_{s,xx})\,\mathrm{d}x \\ &= \frac{1}{2}\int_{\mathbb{R}} (\xi''' - \xi') v_s^2\,\mathrm{d}x. \end{aligned} \tag{C.1}$$

We first take smooth compactly supported densities and then approximate finite signed measures by truncation and convolution. The approximating measures have uniformly bounded

and uniformly tight total variations, and their potentials converge in $\mathcal{H}$. The differentiated kernel is bounded and continuous, since the apparent diagonal jump is multiplied by $\xi(x)-\xi(y)$. Thus (C.1) also holds for the present signed measures. In particular, $|Q_s'| \le 2(L_1+L_3)Q_s$.

The curve v_s is differentiable in $L^2(\mathbb{R})$: it is a Bochner integral of the translated Green kernel against a finite signed measure, and translations of this kernel are C^1 in L^2. The time-dependent weights and shifts have locally bounded derivatives, so differentiation under the integral is justified. For $g \in L^2$ with $\|g\|_2 \le 1$, let $w = (-\partial_x^2 + 1/4)^{-1} g$. Fourier multipliers give $\|w\|_2 \le 4$, $\|w_x\|_2 \le 1$, $\|w_{xx}\|_2 \le 1$. Distributional duality gives

$$\langle \partial_s v_s, g\rangle = \langle v_s, \xi w_x - \tfrac{1}{2}\xi' w\rangle_{\mathcal{H}}.$$

The derivative and zeroth-order bounds for the second argument imply

$$\|\xi w_x - \tfrac{1}{2}\xi' w\|_{\mathcal{H}} \le \tfrac{3}{2}L_0 + \tfrac{3}{2}L_1 + 2L_2 \le 2(L_0+L_1+L_2).$$

Hence $\|\partial_s v_s\|_2 \le 2(L_0+L_1+L_2)\|v_s\|_{\mathcal{H}}$. Differentiating the right-hand side of (C.1) in L^2 now yields

$$Q_s'' = \int_{\mathbb{R}} (\xi''' - \xi')\, v_s\, \partial_s v_s \,\mathrm{d}x, \qquad |Q_s''| \le 4(L_1+L_3)(L_0+L_1+L_2)Q_s.$$

We have therefore obtained the quadratic-form inequalities $-\kappa_i T_m \le T_0^{(i)} \le \kappa_i T_m$. To obtain the asserted factorization, define the symmetric form on $\operatorname{ran} T_m^{1/2}$ by $\mathfrak{b}(T_m^{1/2} f, T_m^{1/2} g) = \langle f, T_0^{(i)} g\rangle$. The diagonal bound and polarization show that this form is bounded with norm at most κ_i. We extend the form to the closure of the range, then extend its representing operator by zero on the orthogonal complement. This proves the lemma. □

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Data availability

The results are theoretical. The manuscript contains all proofs of the new multipeakon estimates. The global solution theory used here is stated explicitly and proved in Part I [22]. The complete Part I manuscript, its LaTeX sources, figures and numerical supplement are supplied in companion_materials/Part_I/. The LaTeX sources of the present paper also accompany this submission.

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